

Data-Driven Diagnosis: A Machine Learning Study on Cognitive Impairment



Presented by:-

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Introduction:

- Mild Cognitive Impairment (MCI) is a precursor stage of Alzheimer's disease (AD) that affects memory, cognition, and behavior.
- About 15-20% of people with MCI are estimated to develop AD within 5 years of diagnosis.
- Current diagnostic methods for MCI and AD have limitations in terms of accuracy and accessibility.
- Gene expression profiling has emerged as a promising approach for identifying molecular biomarkers that can differentiate between healthy controls, MCI patients, and AD patients.
- Machine learning algorithms can be trained on these biomarkers to classify patients based on their gene expression values, which can provide a more accurate and accessible diagnosis for MCI and AD.

Background study:-

- Shu et al. (2021) explores the prediction of progression from mild cognitive impairment (MCI) to Alzheimer's disease (AD) using a radiomics-integrated model. **MCI is widely considered a prodromal stage of AD**, and the study highlights the need for improved diagnostic methods in this population.
- Albert et al. (2011) provides recommendations for diagnosing mild cognitive impairment (MCI) due to Alzheimer's disease (AD). This paper highlights the significance of **MCI as a prodromal stage of AD**, emphasizing the importance of early detection and intervention.
- Cui et al. (2018) conducted a study on exercise intervention and cognitive improvement in Alzheimer's disease (AD). The study emphasizes the importance of early intervention, as **mild cognitive impairment (MCI) is considered a preclinical stage of AD**.



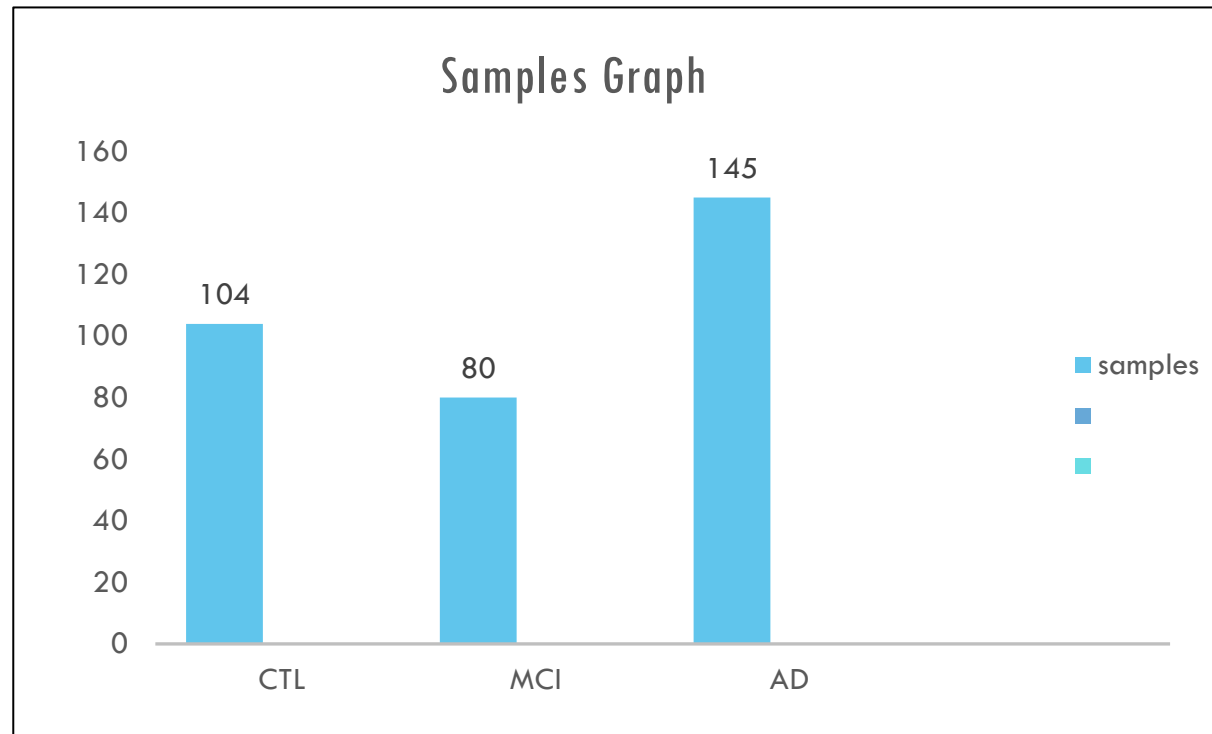
Objectives:-

- Develop an accurate and robust machine learning model that is precise and reliable in predicting the stages of individuals as Alzheimer's disease (AD), Mild Cognitive Impairment (MCI), or healthy, using gene expression data.
- Investigate the potential of gene expression data in distinguishing between AD, MCI, and healthy stages, and identify the key genes or gene expression patterns that contribute to the classification, providing insights into the molecular mechanisms underlying the progression of AD and MCI.

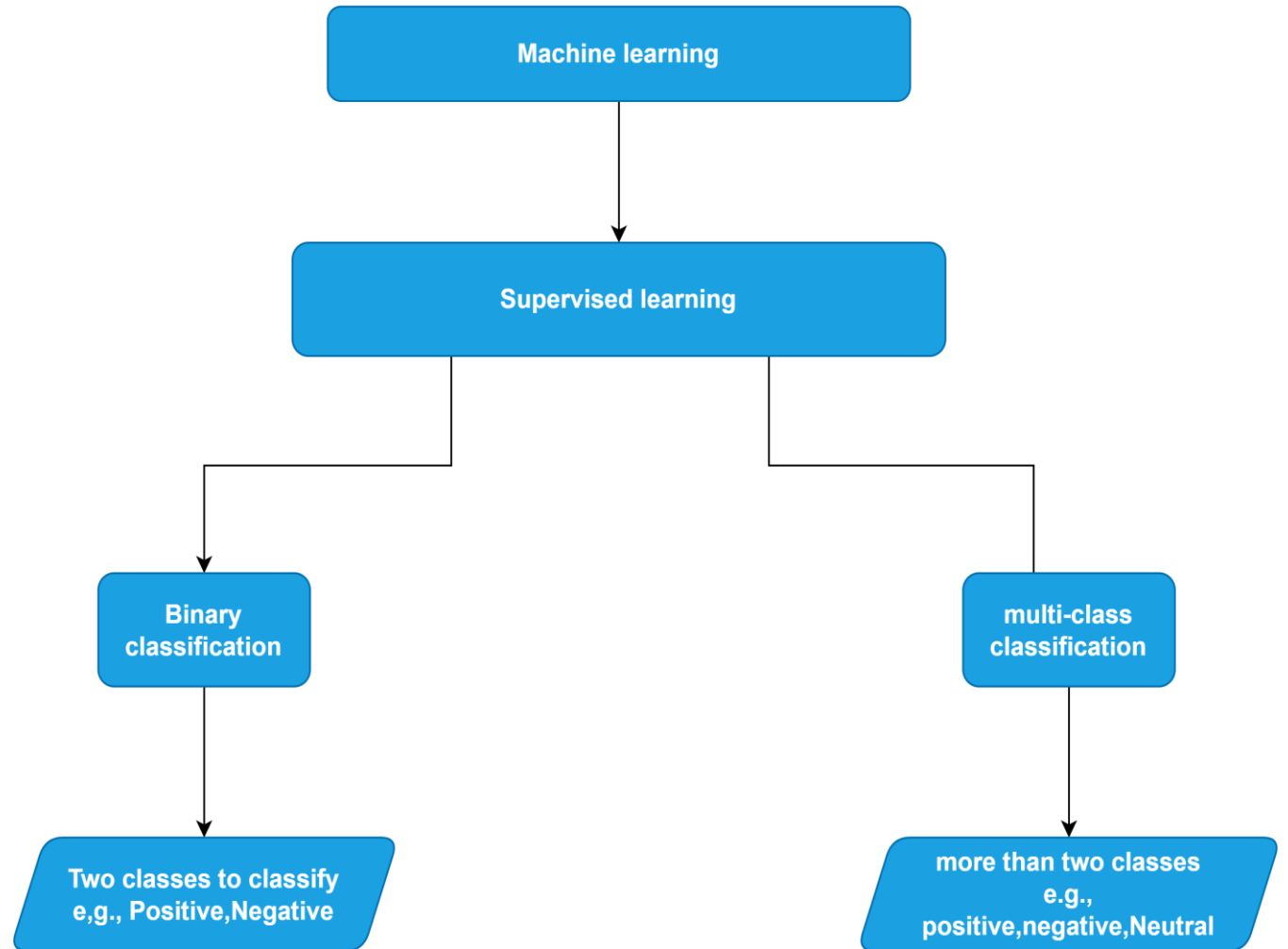
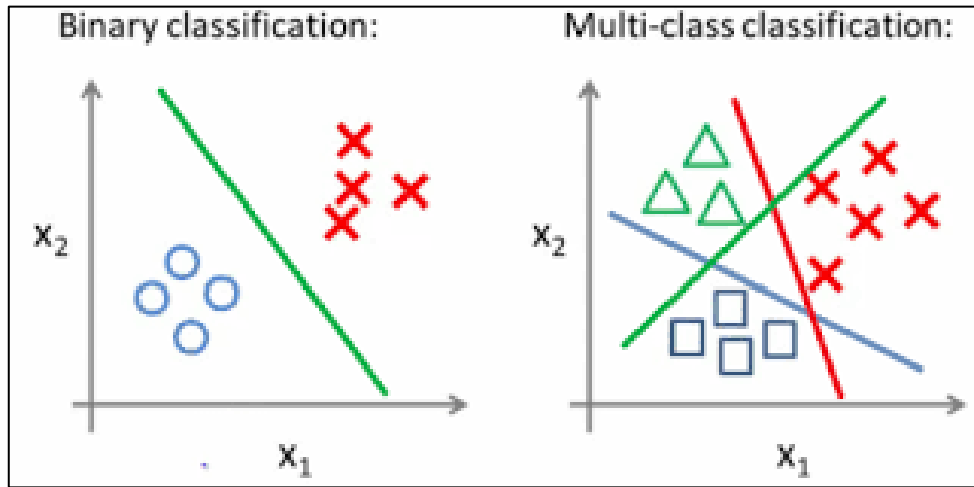
Materials & Methods:-

Dataset:-

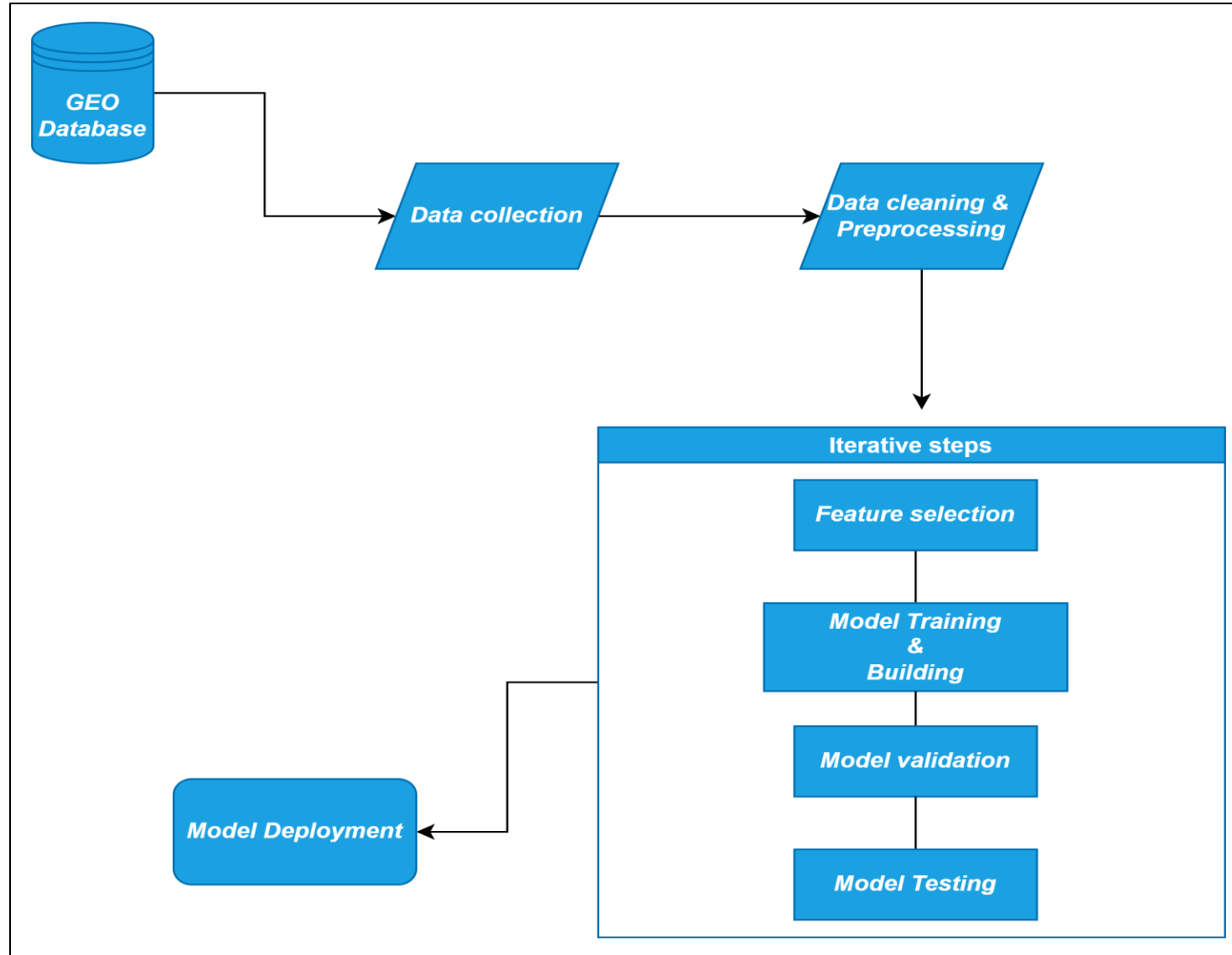
- The dataset for this study was downloaded from the NCBI, Gene expression Omnibus (GEO) using [GSE63063](#) accession number. The platform of the chip analyzer was [GPL6947](#).
- Dataset was narrowed down to 104 control(healthy) samples including 80 samples of MCI, and 145 samples Alzheimer's disease cases. The samples summed up in 329 samples, including 200 female and 129 male patients.



Binary & Multiclass:-



Methodology:



Experimental Trails:-

- Trail_1:- features selected by studying literature.
- Trail_2 :-Feature selection using GEO2R tool and considering p-value & logFC values
- Trail_3 :- Feature selection using GEO2R tool this time with adj-p-values and LogFC values.

Trail 1:-

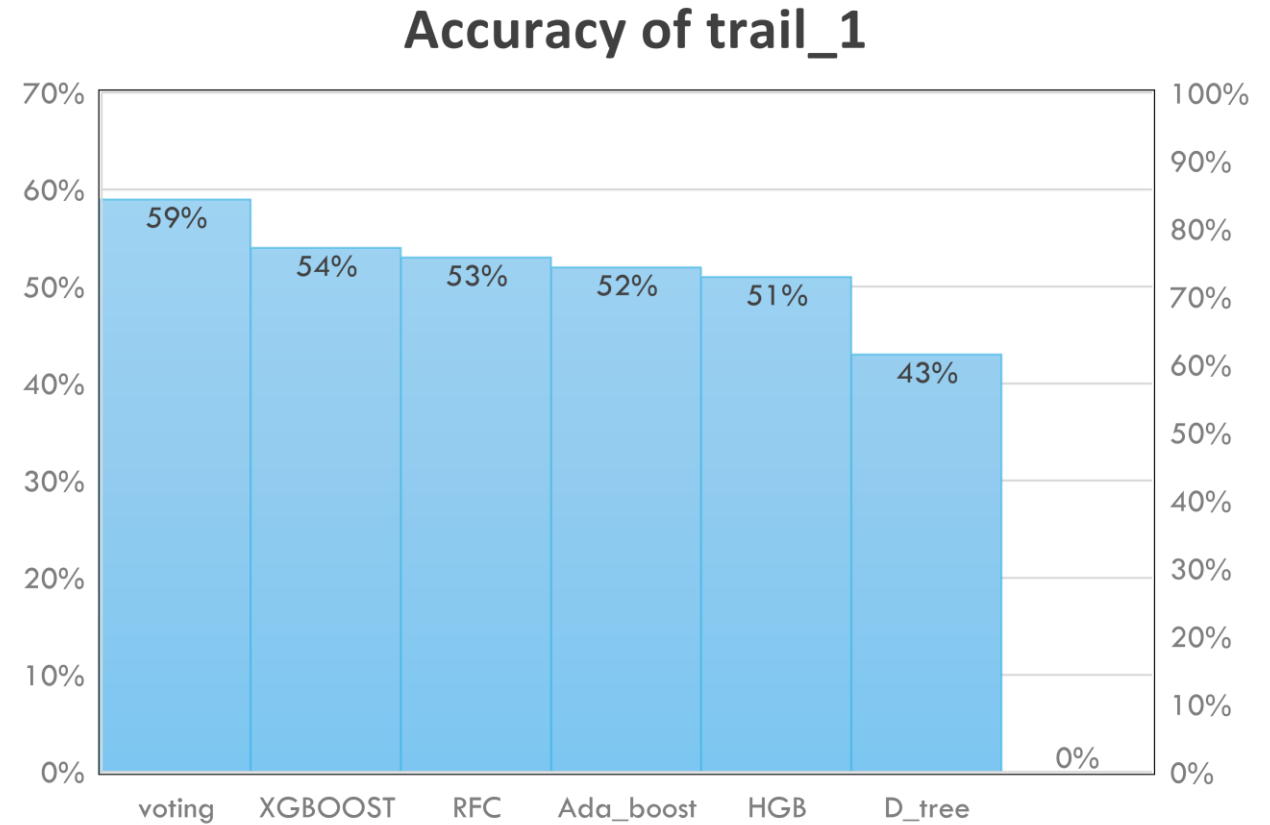
- ❑ The article by Soleimani Zakeri et al. presents a gene co-expression network approach for gene biomarker discovery in different stages of Alzheimer's disease (AD). This article represents the work on same GEO Dataset which selected for case study and the genes discovered here are used as features for machine learning classification.

The next group involved the genes which belonged to the overlap between MCI-AD module groups and CTL-AD module groups. There were eight genes in this group, including ALDOC, APBA3, CHST14, DDX19A, HNRNPUL1, KCTD2, LAMTOR1, and RPA1. The first one, ALDOC, was found to be related to Alzheimer's disease in a study that investigated the proteomics in Alzheimer's brain⁴⁹. Another study, which discussed this gene, examined AD pathogenesis and its related key regulators⁵⁰. The second gene, APBA3, and its interaction with beta-amyloid highlight the importance of the examination of its genomic structure⁵¹.

In These total of **27** genes are identified as Biomarkers DEG's which are used as features in trail_1

Results of Trail 1:-

After features selection extracted data used to train the model and validated with test data and the model judged on different metrics like **Accuracy, precision,F1,Recall**



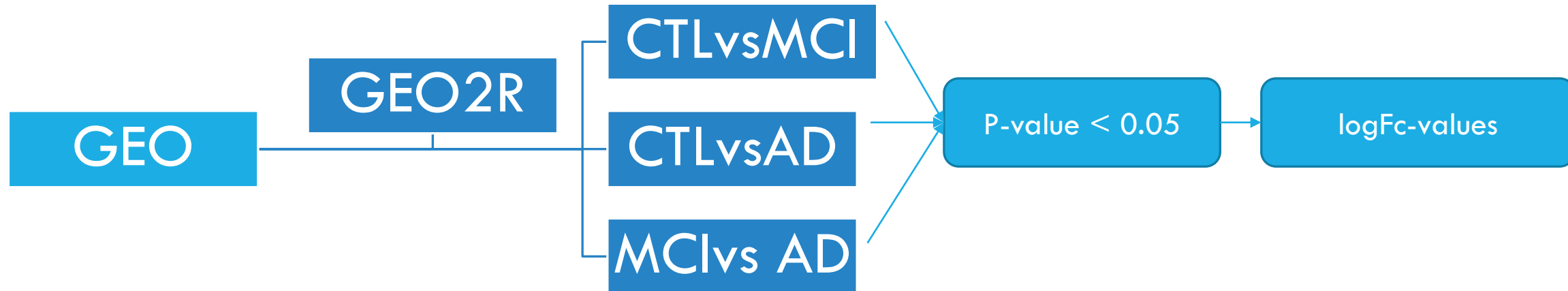
Trail_2:-

In this trail features (genes) are selected after analysing gene expression data with GEO2R tool.

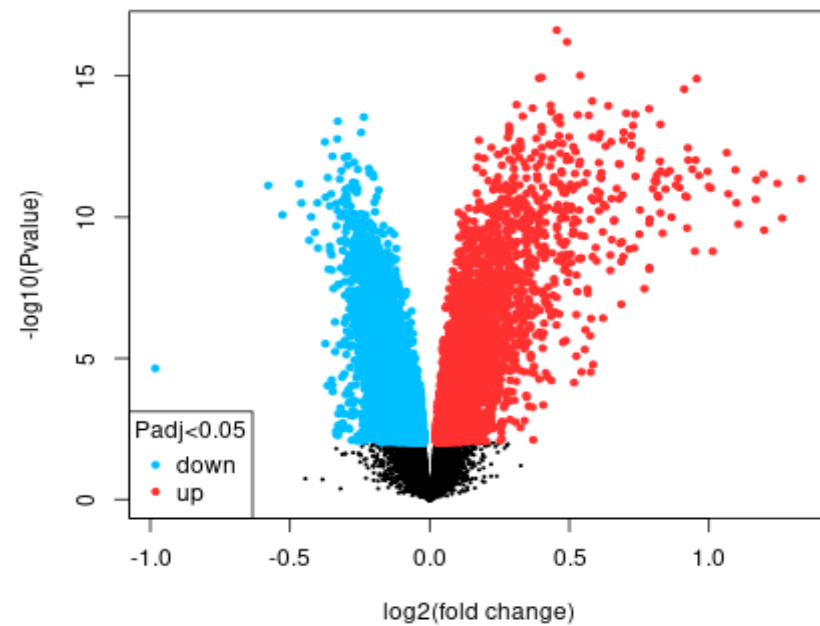
The GEO2R tool was analysed for three times for three different conditions.

- ☐ CTL_vs_MCI
- ☐ CTL_vs_AD
- ☐ MCI_vs_AD

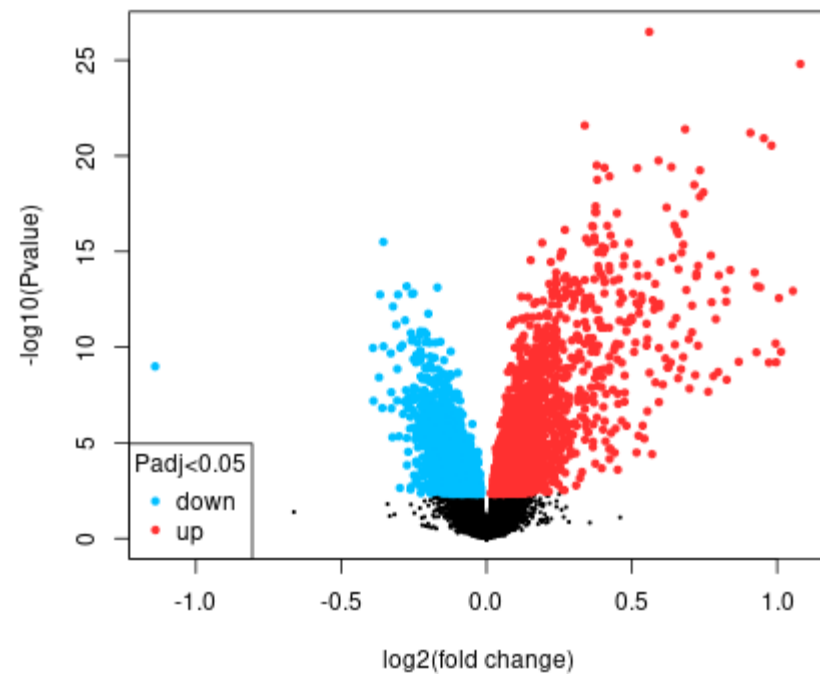
GEO2R is inbuilt tool in **GEO** which works on 'LIMMA' package one of the library of R-programming language.



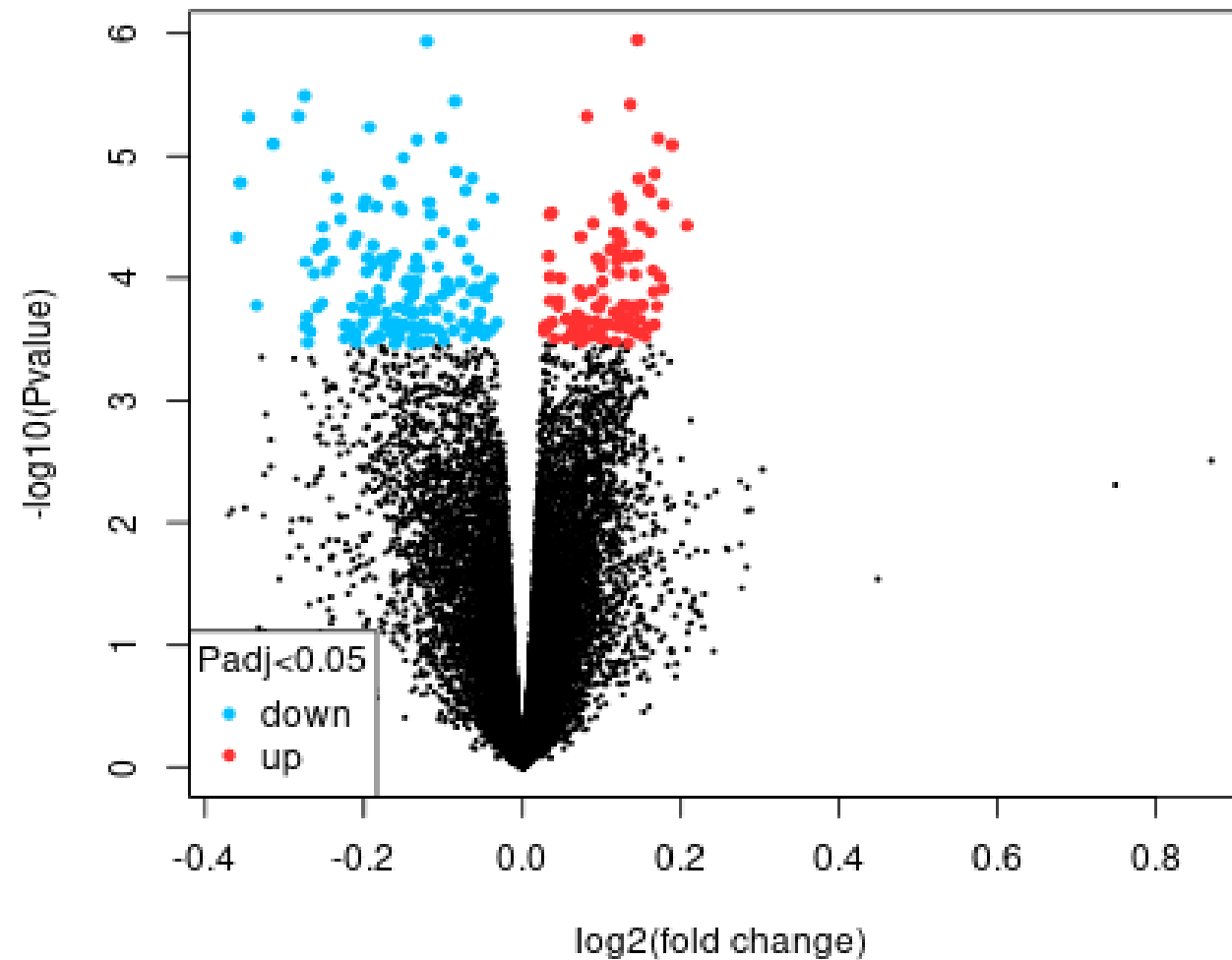
GSE63063: CTL vs MCI



GSE63063: CTL vs AD



GSE63063: MCI vs AD



What is P-value?

In statistical hypothesis testing, p-value is used as a measure of the strength of evidence against the null hypothesis. In gene expression analysis, p-value can be used to select differentially expressed genes. For example, if we are comparing the gene expression levels between two conditions (e.g. disease vs control), we can calculate the p-value for each gene using a statistical test such as t-test. A low p-value (e.g. less than 0.05) indicates that the difference in expression levels between the two conditions is statistically significant, and the gene is likely to be differentially expressed.

Summary

$p < 0.05 \rightarrow$ statistically significant difference

$p > 0.05 \rightarrow$ no statistically significant difference

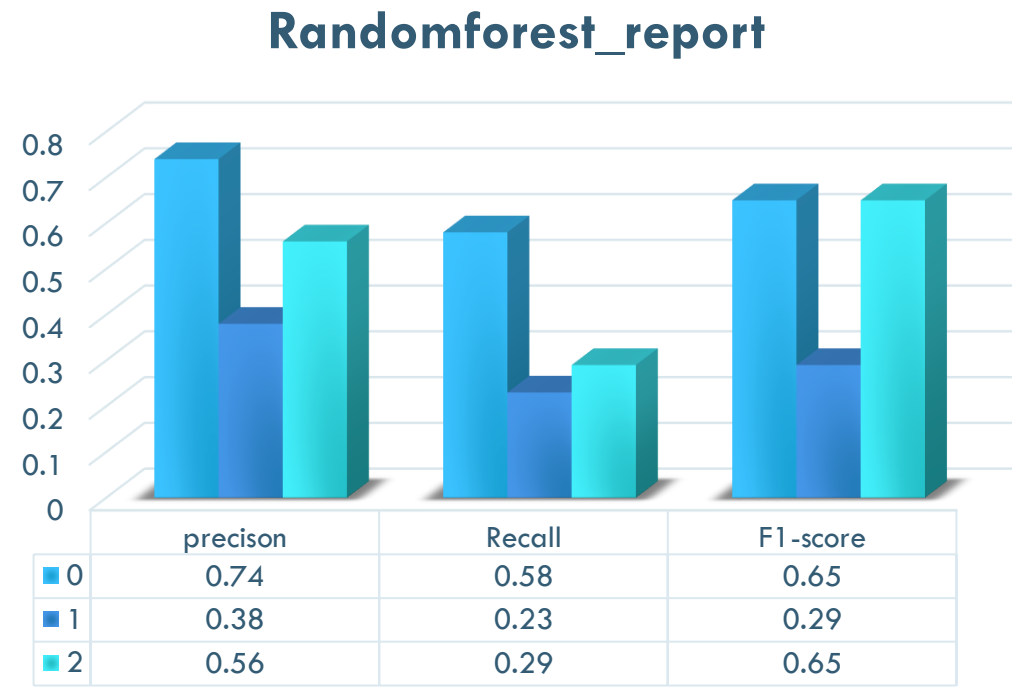
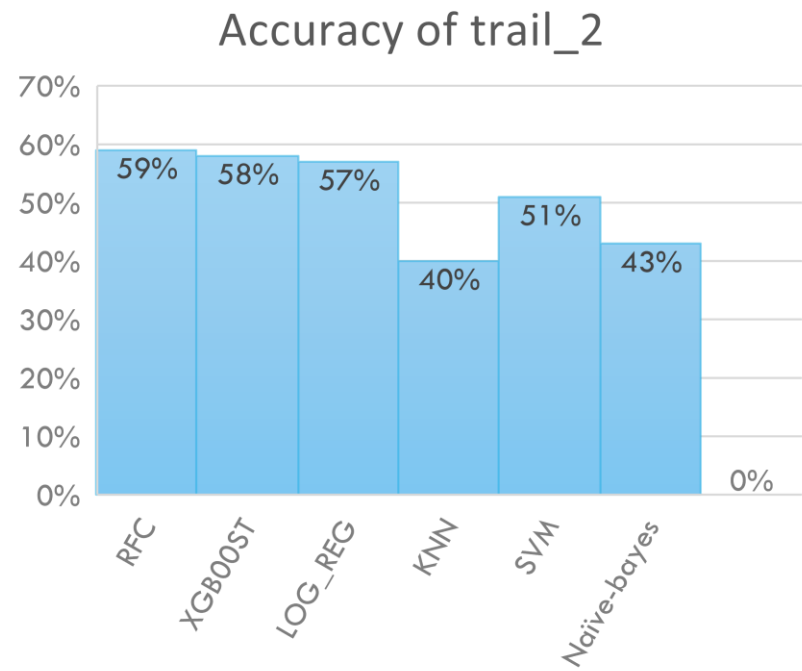
Logfc values:-

- logFC (log fold change) is a measure of the difference in expression levels between two conditions, usually treated and untreated, control and experimental, or different stages of a disease.
- A logFC of zero means that there is no difference in expression levels, while a positive or negative logFC indicates an up- or down-regulation of the gene in the experimental condition compared to the control.

logFc Value	in context of gene expression
Positive	Upregulated
If equals = 0	No difference in expression
negative	Downregulated

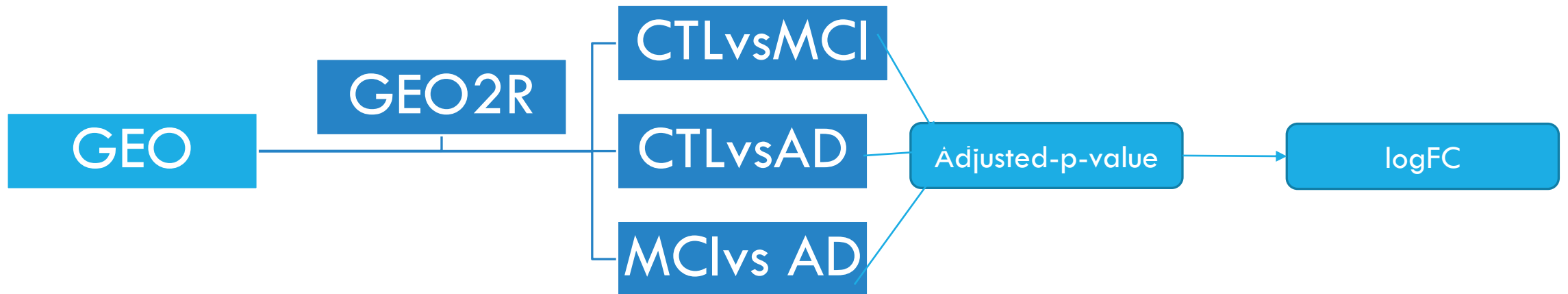
Results of trail_2:-

Using $p\text{-value} < 0.05$, selected total of 300 genes of 100 of each condition(i.e, CTL vs MCI) were 50 upregulated & 50 down regulated and by filtering all the mutual genes in three conditions.



Trail_3:-

In trail_3 choosing of features was done by considering 'adjusted-p-values' followed by logFc values.



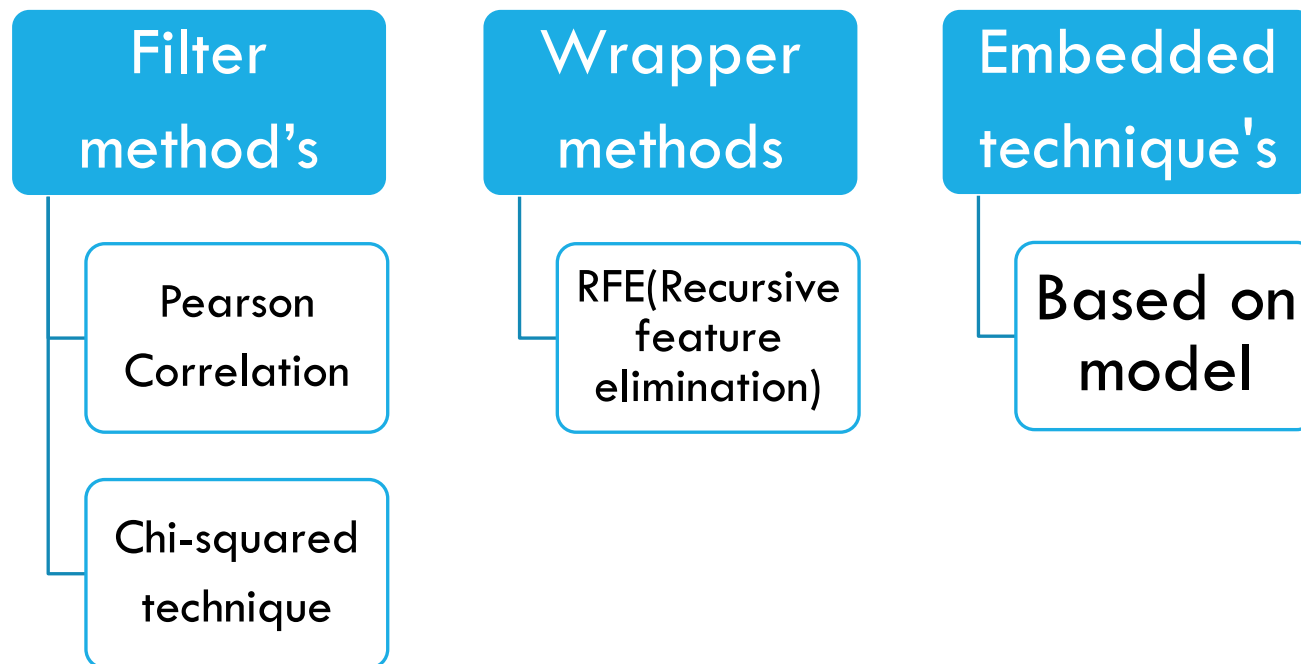
Adjusted-p-value:-

The **adjusted p-value** is a modified version of the p-value that takes into account the number of tests performed. A common method for adjusting the p-value is the **Benjamini-Hochberg** procedure, which controls the **false discovery rate** at a predetermined level.

For example, let's say we are comparing gene expression levels between two groups, and we perform a t-test on the expression values for each gene. We obtain a p-value of 0.01 for gene A, which suggests that the gene is differentially expressed between the two groups. However, we have tested 10,000 genes, so the chance of a false positive is high. If we adjust the p-value using the Benjamini-Hochberg procedure with a false discovery rate of 0.05, we might find that the adjusted p-value for gene A is now 0.1, which is not significant at the adjusted level. This means that we should not conclude that gene A is differentially expressed between the two groups.

Feature selection

- In trail_3 we have taken care about feature selection. We have used different feature selection methods and used two classifiers support vector classifier(SVC) & Random Forest classifier(RFC).
- Feature selection involves selecting a subset of the original features from the dataset while discarding the irrelevant or redundant ones. The objective is to identify the most informative and discriminate features that contribute significantly to the predictive power of the model. We have reduced our features to 200 i.e., genes from each feature selection technique



Filter methods:-

1. Pearson correlation:-

This technique is particularly effective in identifying **linear dependencies** between features and the target variable. It performs well when the relationship between features and the target variable follows a linear pattern.

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

r	=	correlation coefficient
x	=	values of the x-variable in a sample
$\bar{x}$	=	mean of the values of the x-variable
y	=	values of the y-variable in a sample
$\bar{y}$	=	mean of the values of the y-variable

2.Chi-square method:-

It evaluates the statistical significance of the association using the chi-squared statistic. This approach is suitable for identifying dependencies between categorical features and the target variable. Unlike Pearson correlation, it can capture associations that are **nonlinear or non-monotonic** in nature.

$$\chi^2 = \frac{\sum O_i - E_i^2}{E_i}$$

χ^2 = Chi-Square value

O_i = Observed values

E_i = Expected values

2. Wrapper methods: -

These methods evaluate different subsets of features by training and testing the model on each subset. This approach considers the performance of the model as the criterion for selecting features.

Recursive Feature Elimination (RFE): -

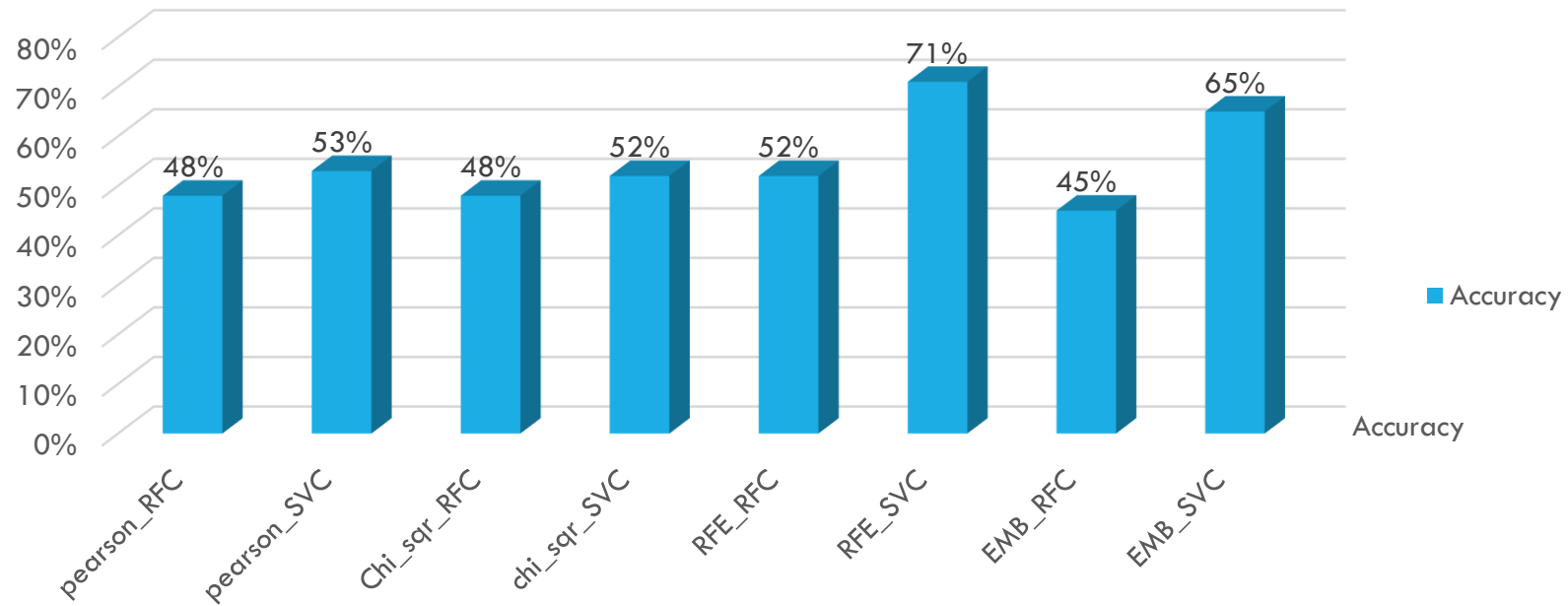
RFE is a technique used to select features based on the **weights assigned to them by an external estimator**, such as the coefficients of a linear model. The process involves recursively considering smaller and smaller sets of features. The objective is to **iteratively identify and retain the most relevant features** based on their assigned weights.

3.Embedded techniques: -

- Embedded methods encompass feature selection within the learning algorithm itself. These techniques select features during the model training process based on their relevance and impact on the overall model performance.
- By integrating feature selection directly into the learning algorithm, the need for separate feature selection steps is eliminated.
- However, embedded methods are **algorithm-specific**, meaning that each algorithm incorporates its own embedded feature selection mechanisms.

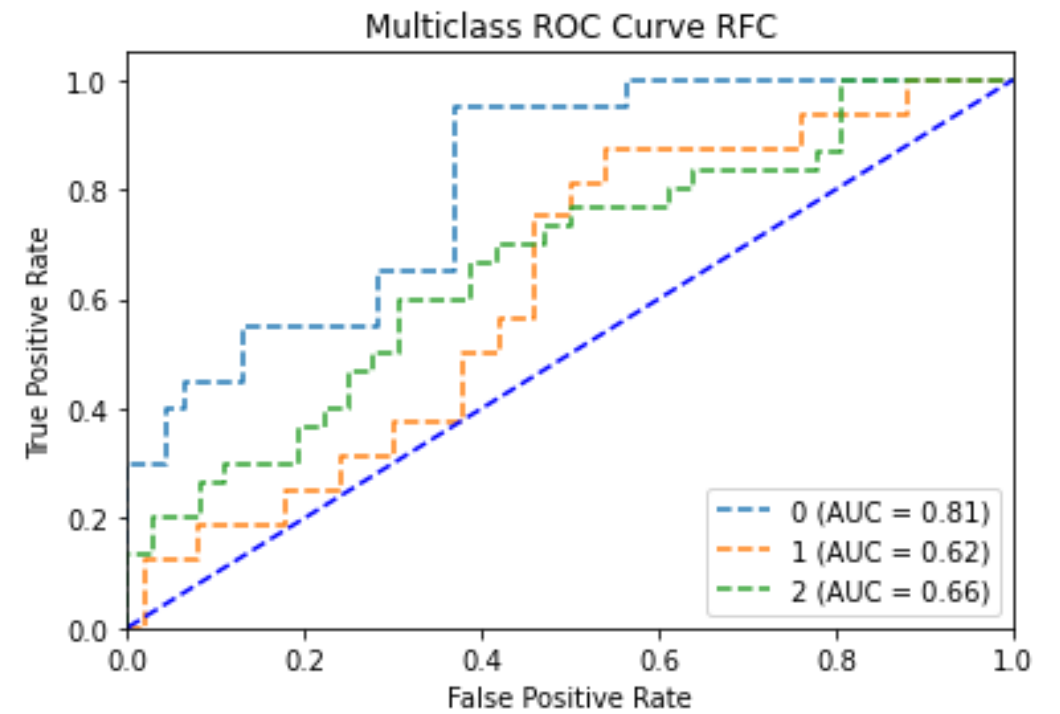
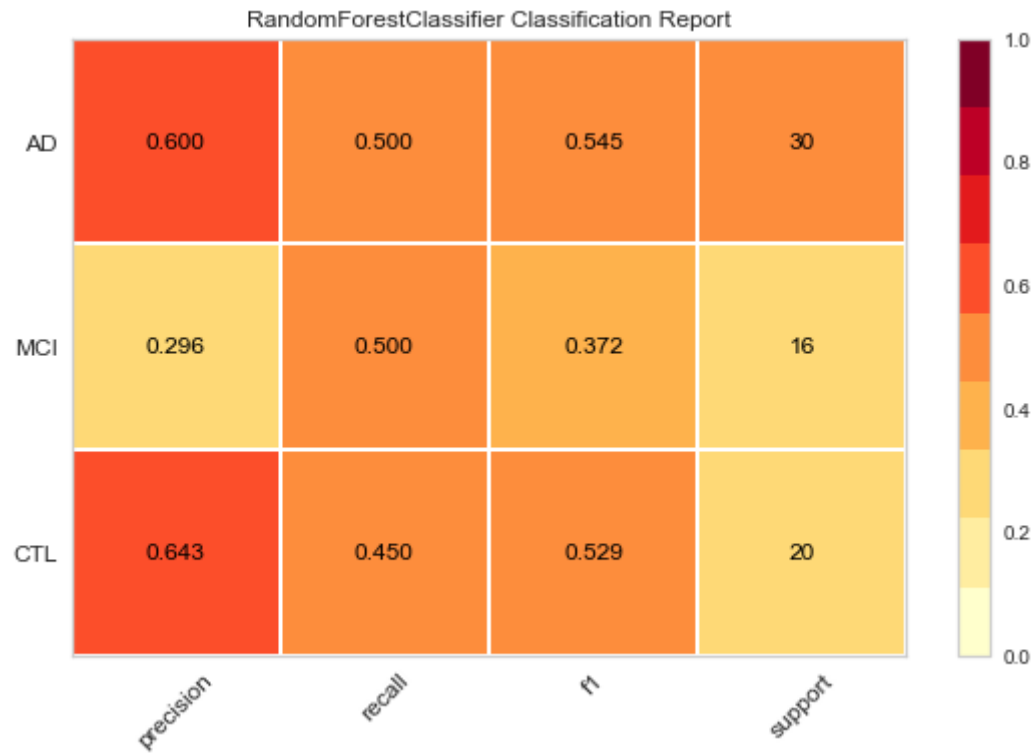
Trail 3 results:-

Accuracy of Trail_3 classifiers

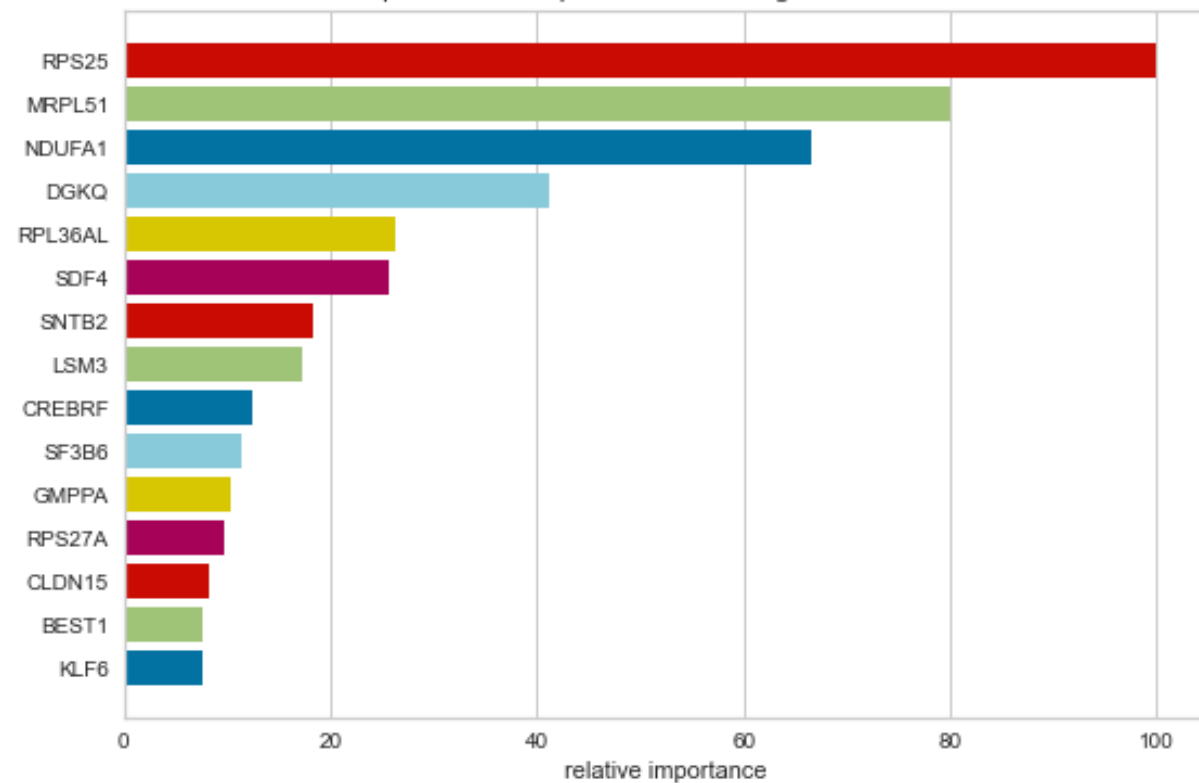


Trail_3_results:-

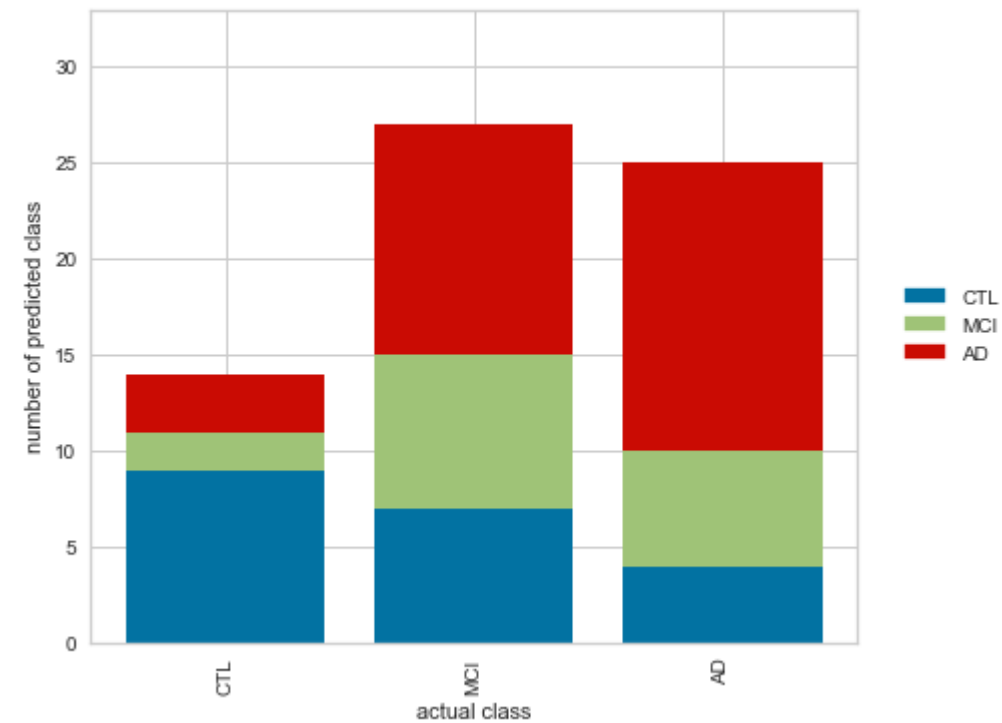
1) Training on Pearson correlation selected features with Random Forest:-



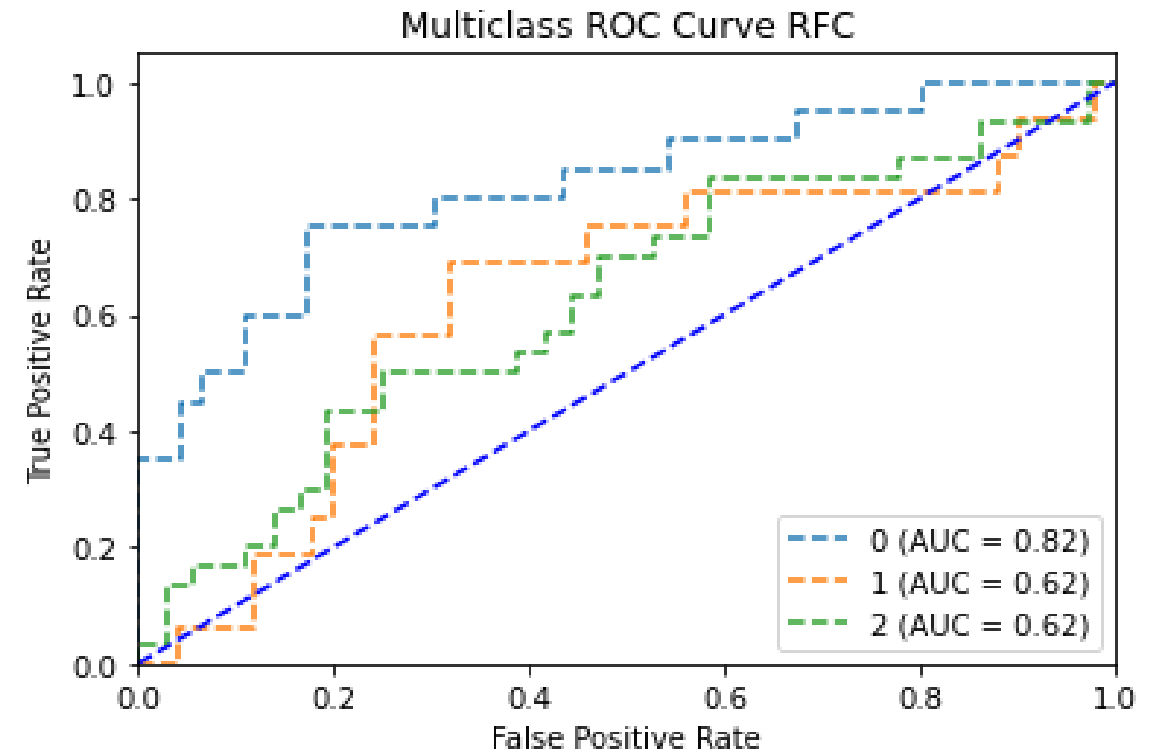
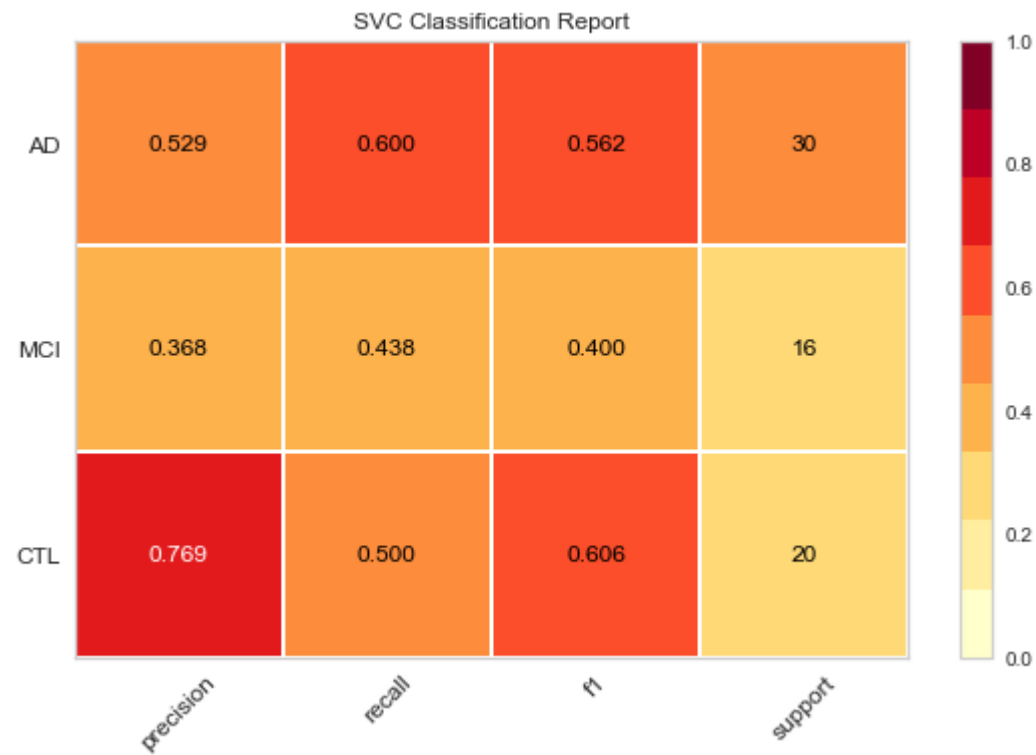
Feature Importances of Top 15 Features using RandomForestClassifier

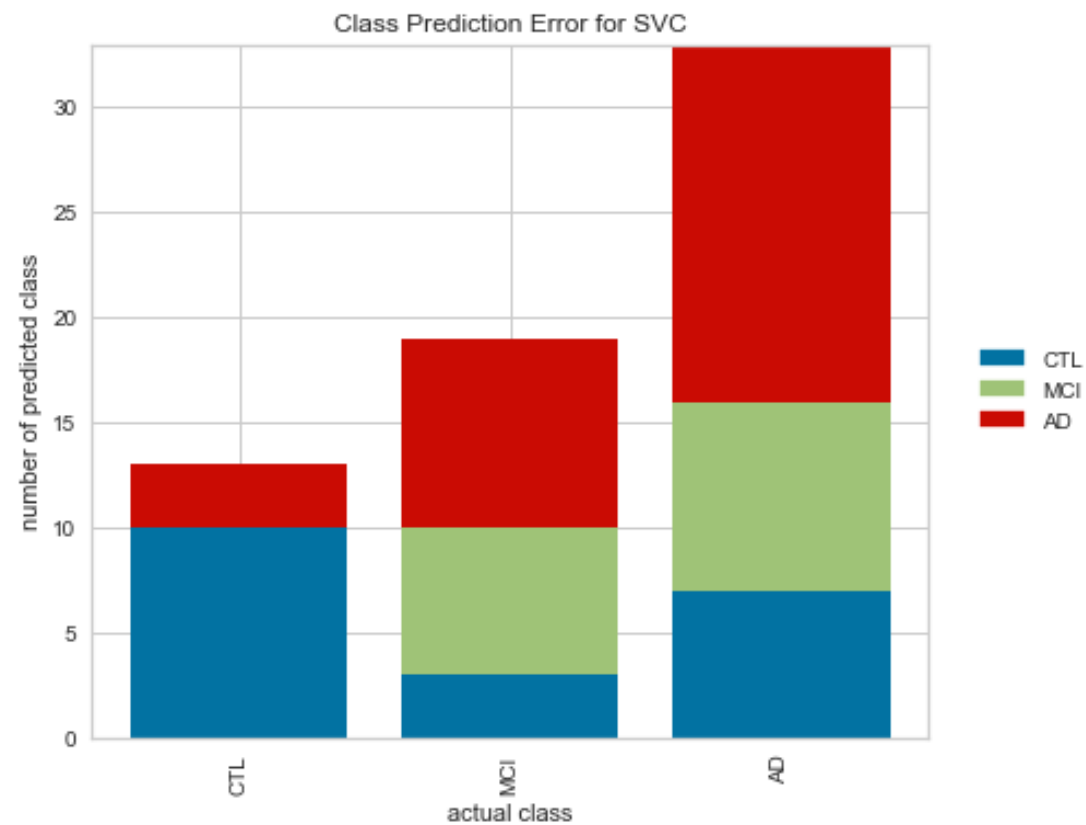
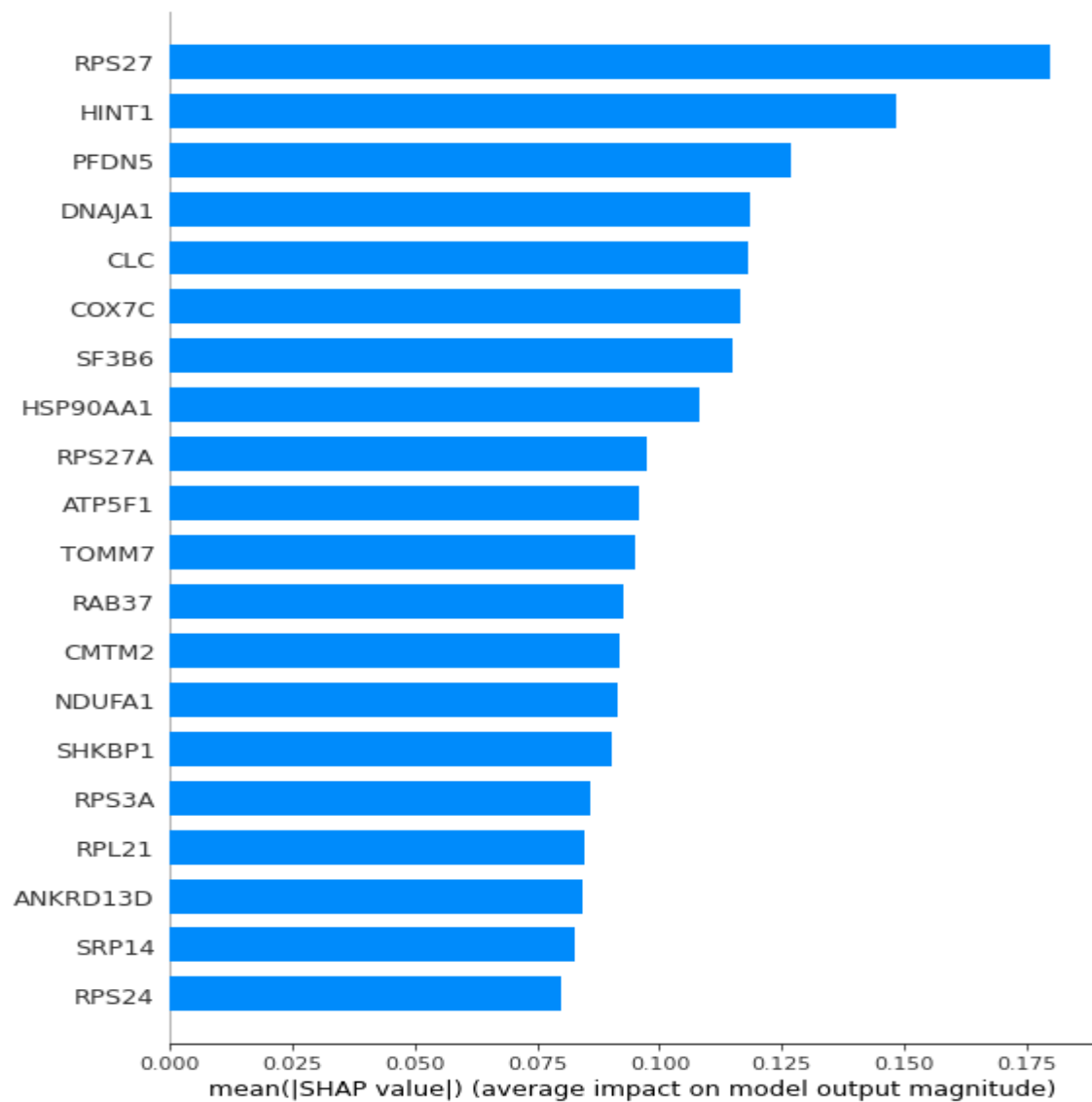


Class Prediction Error for RandomForestClassifier

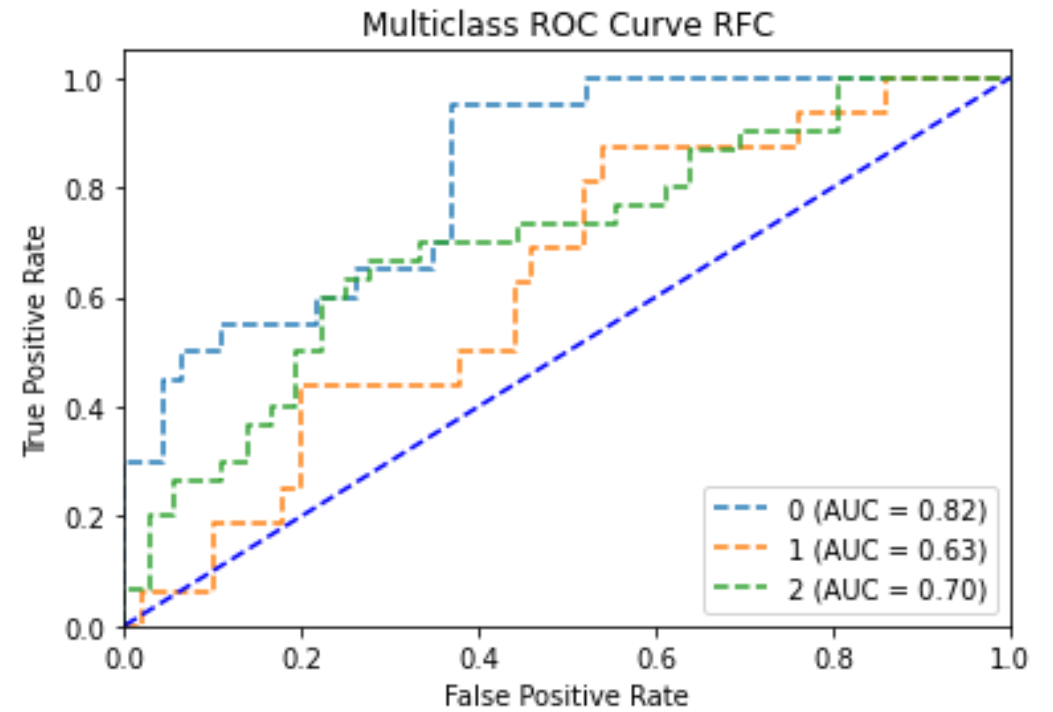
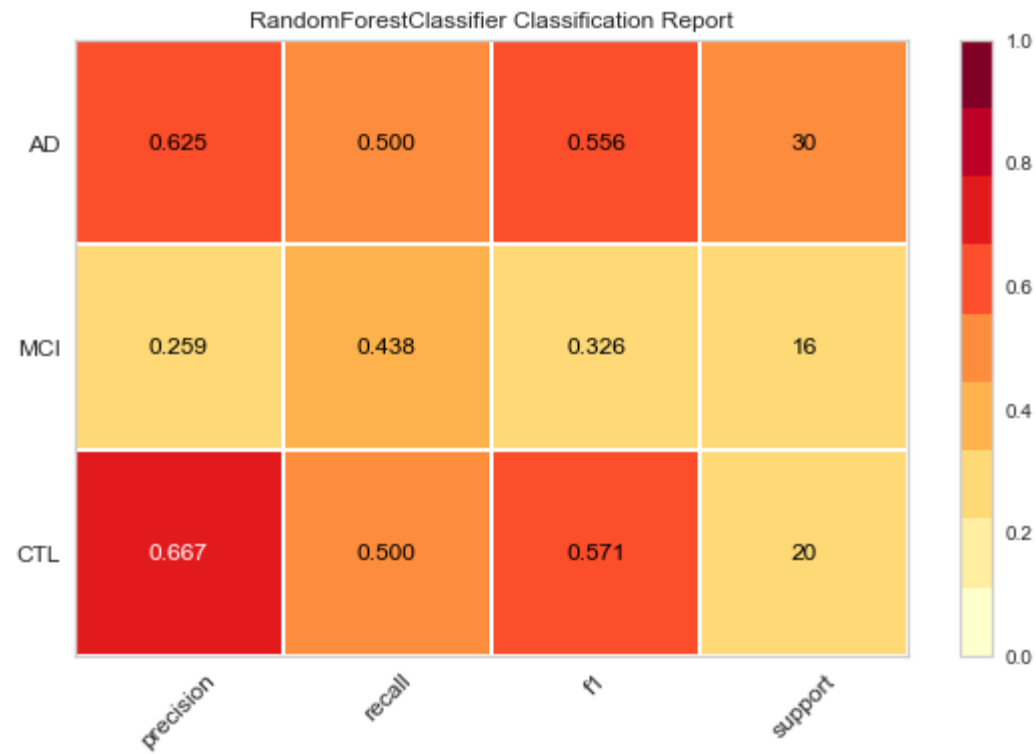


2.Training Pearson selected features on SVC :-

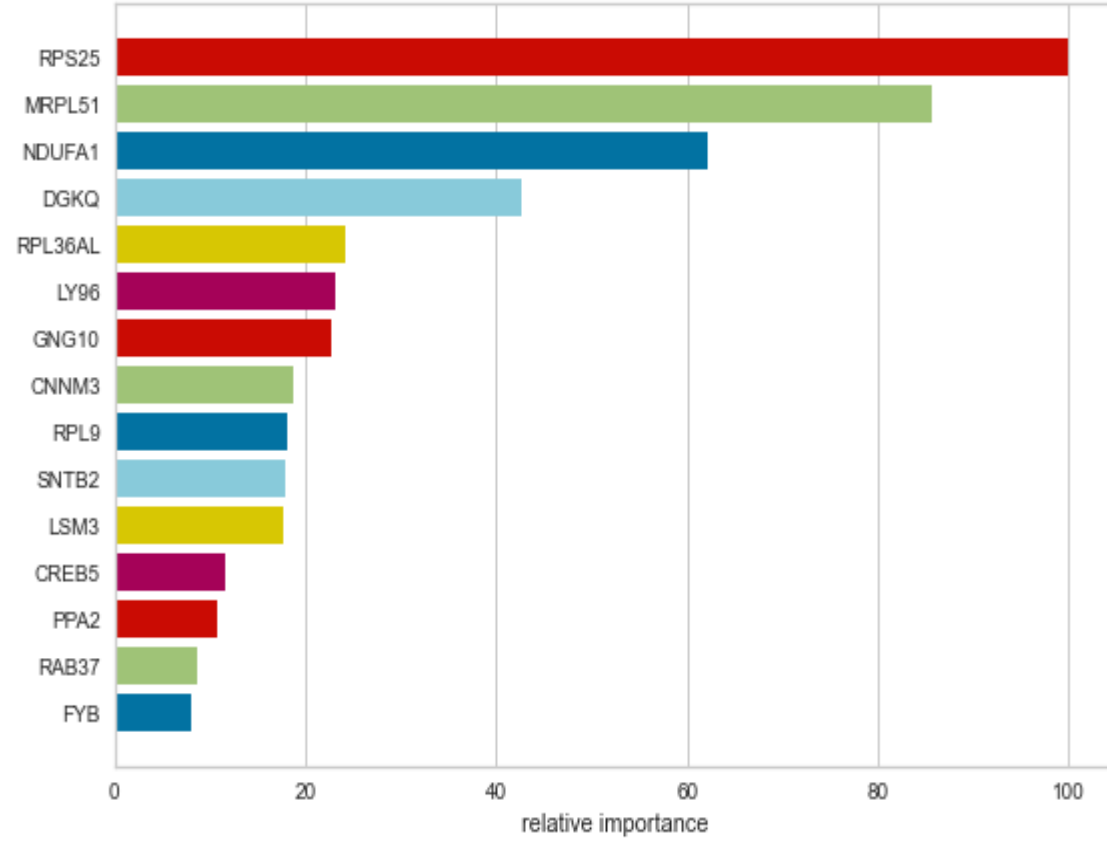




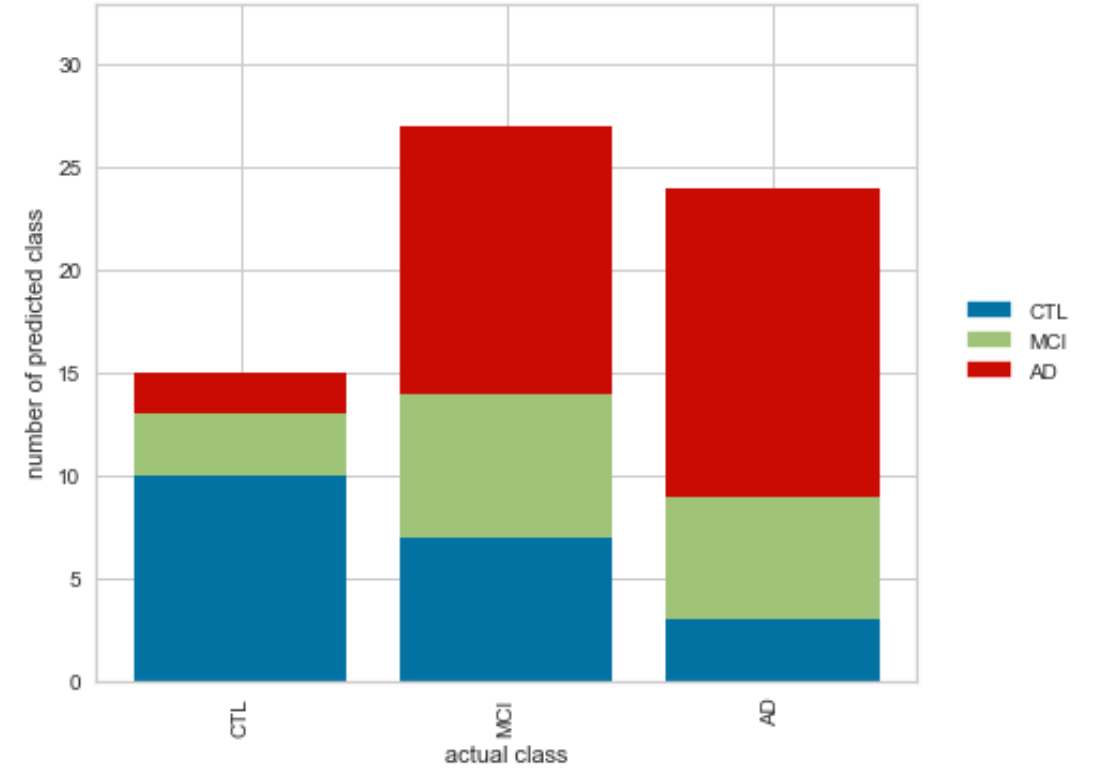
Training model on Chi-square selected features with RandomForest:-



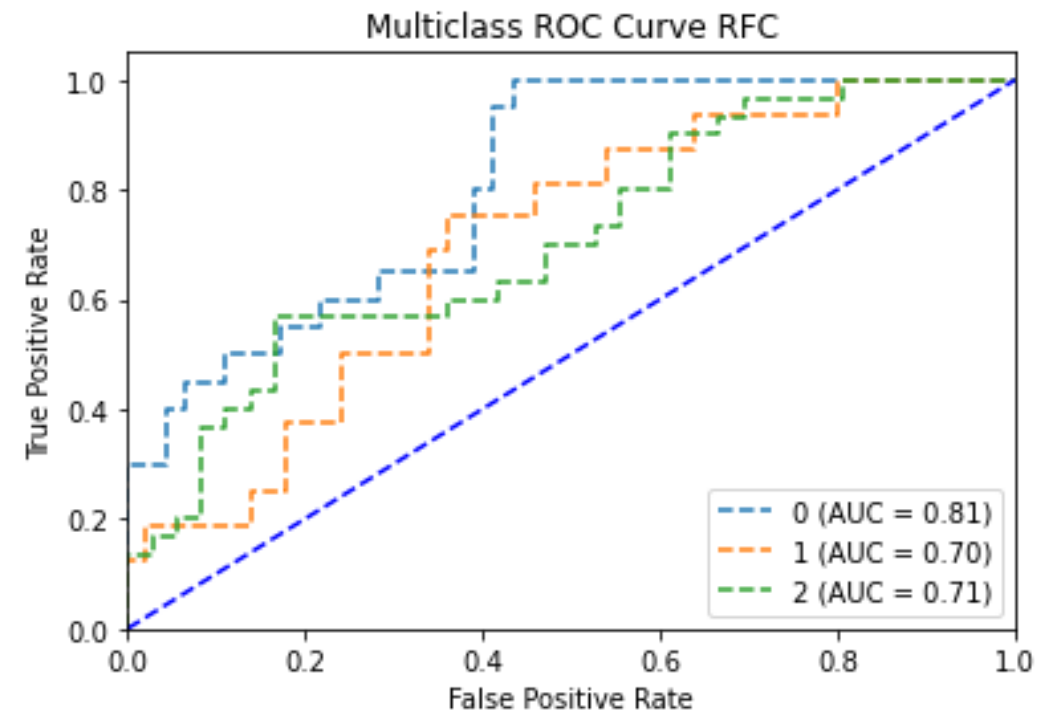
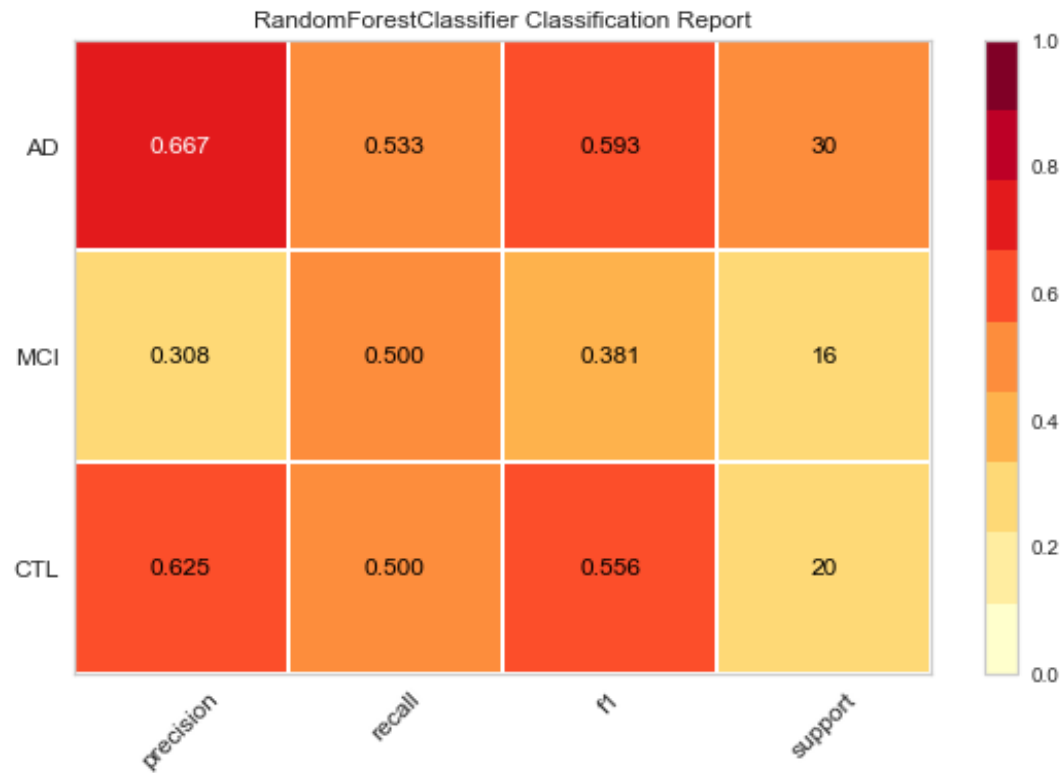
Feature Importances of Top 15 Features using RandomForestClassifier



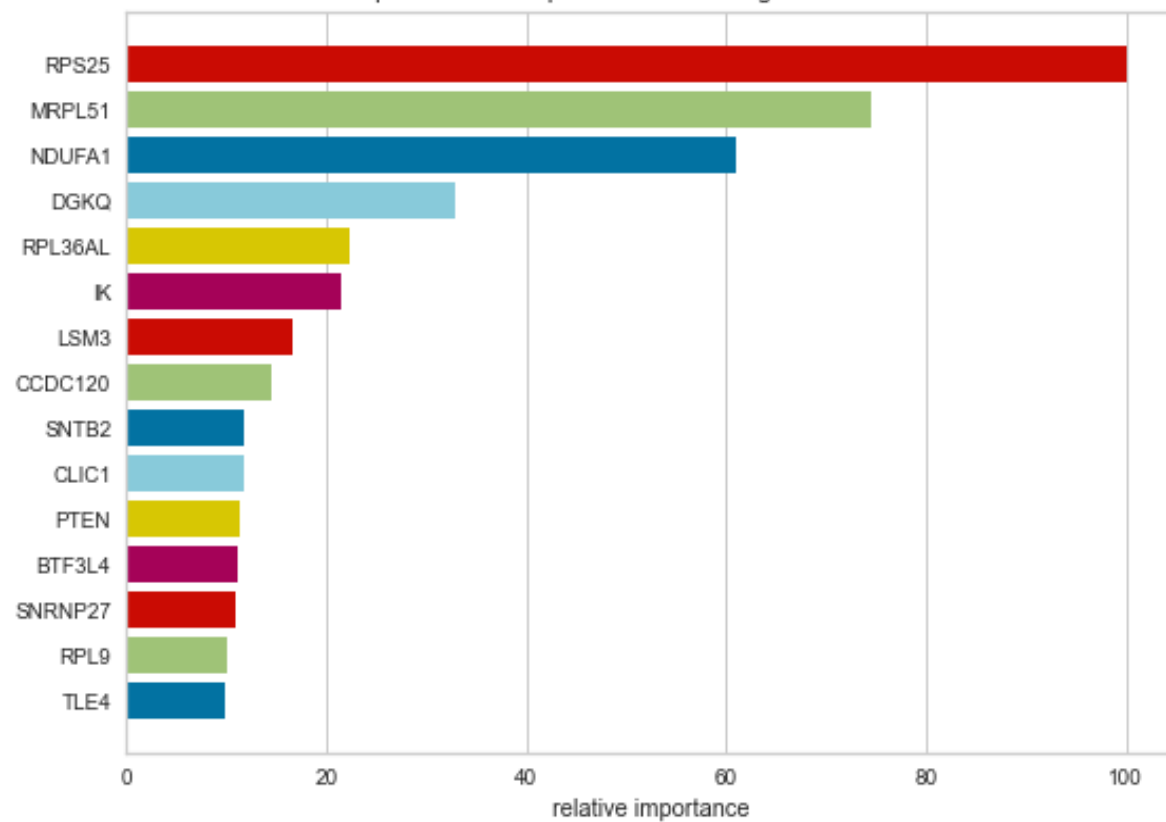
Class Prediction Error for RandomForestClassifier



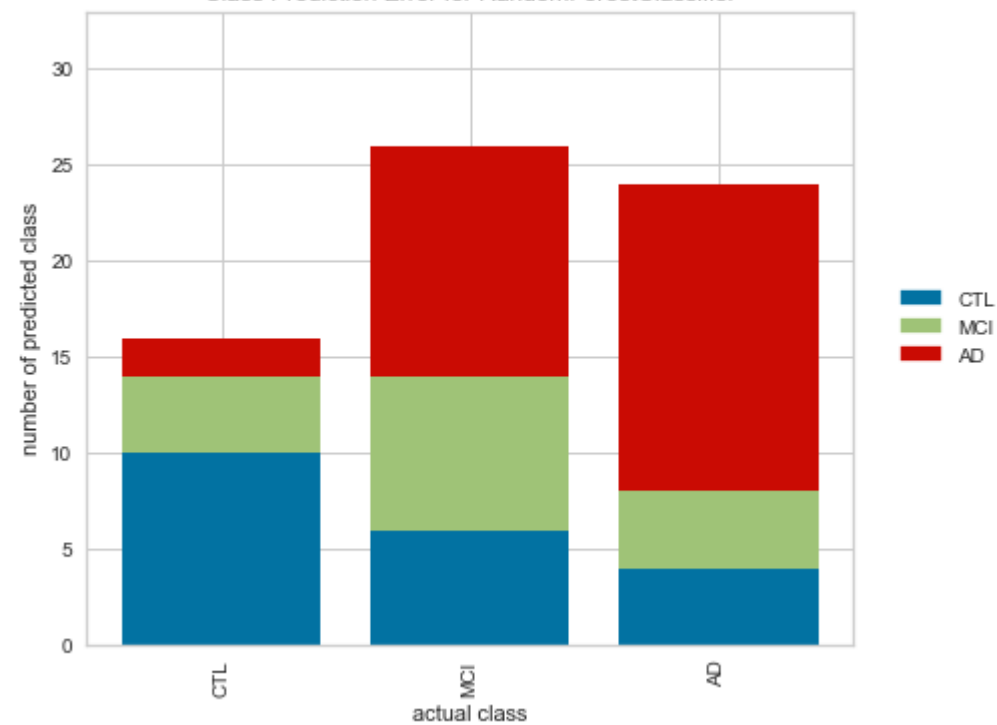
RFE_RF selected features:-



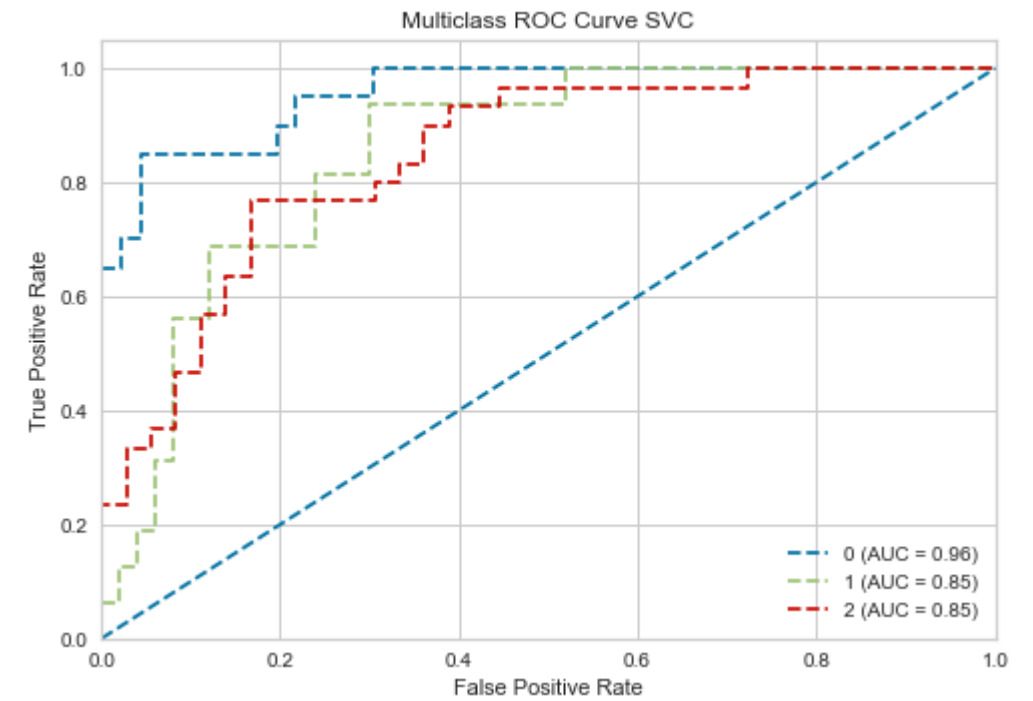
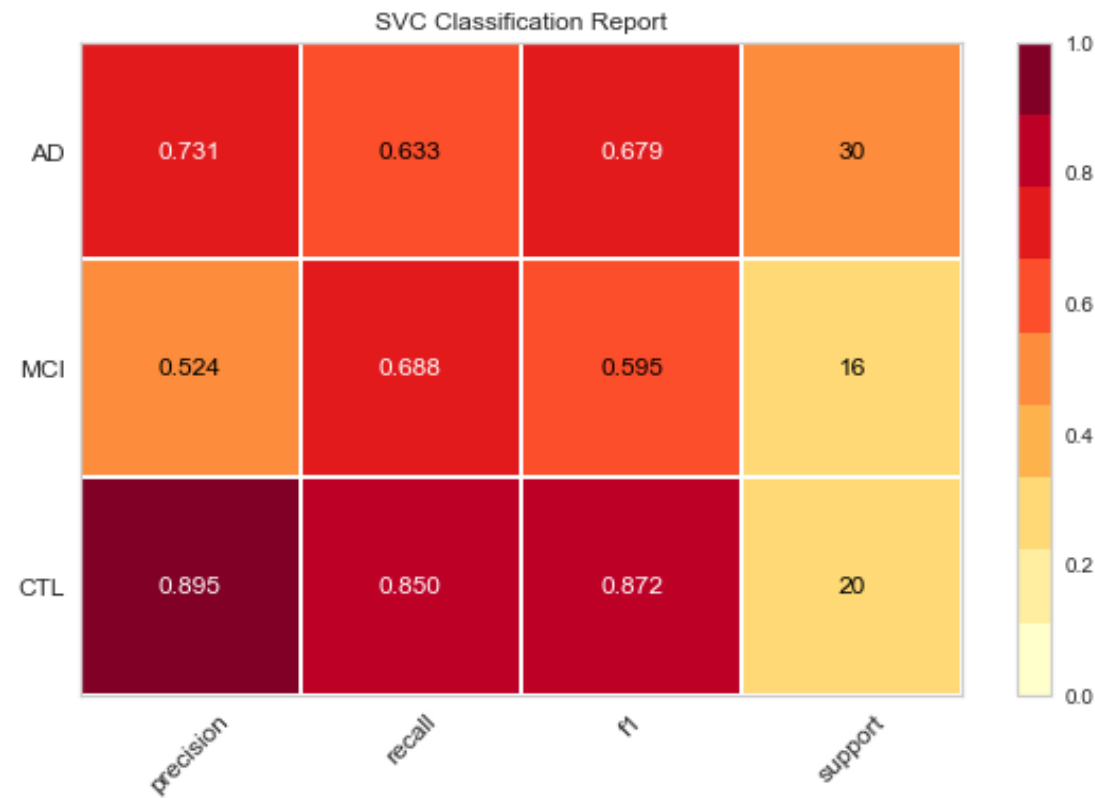
Feature Importances of Top 15 Features using RandomForestClassifier

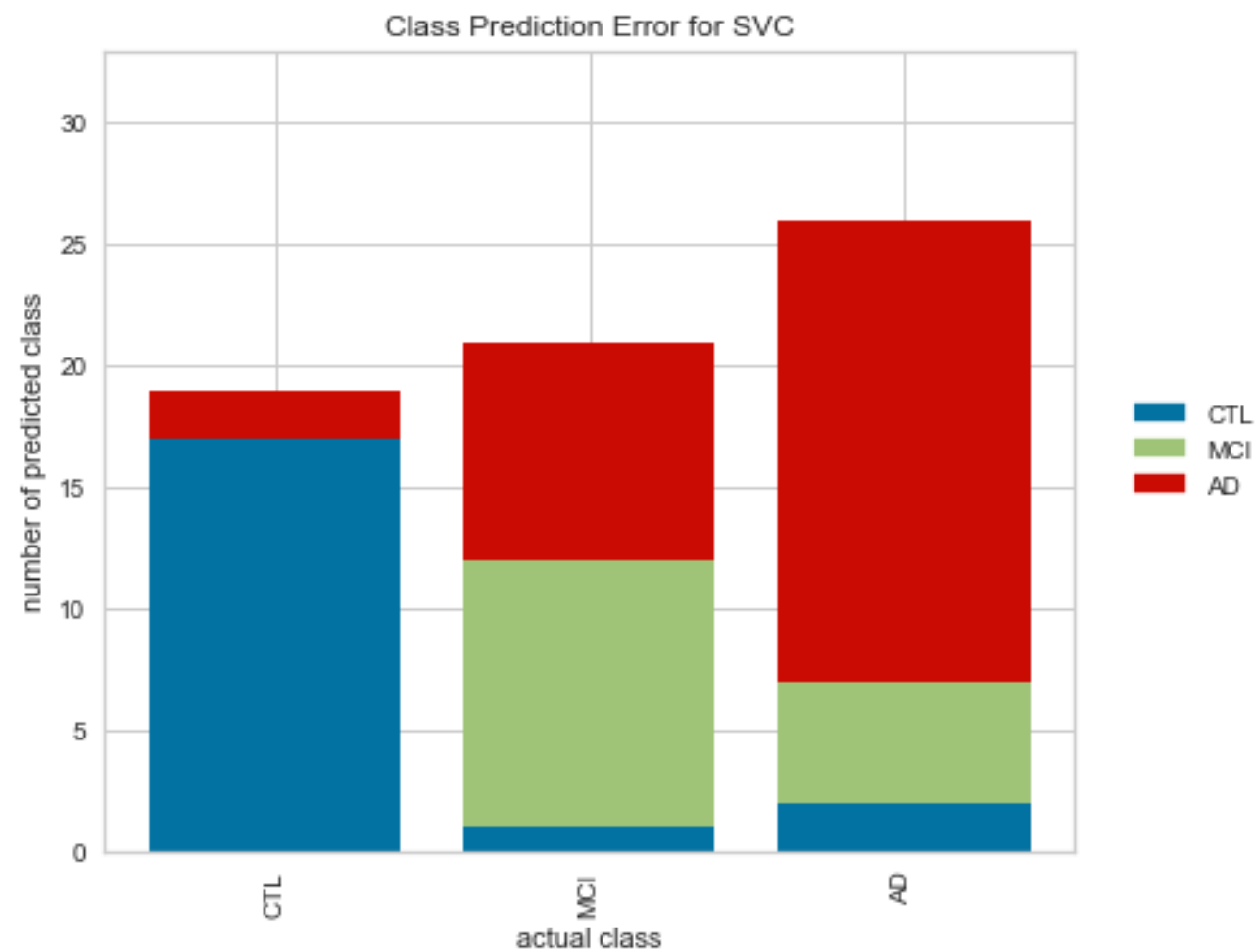
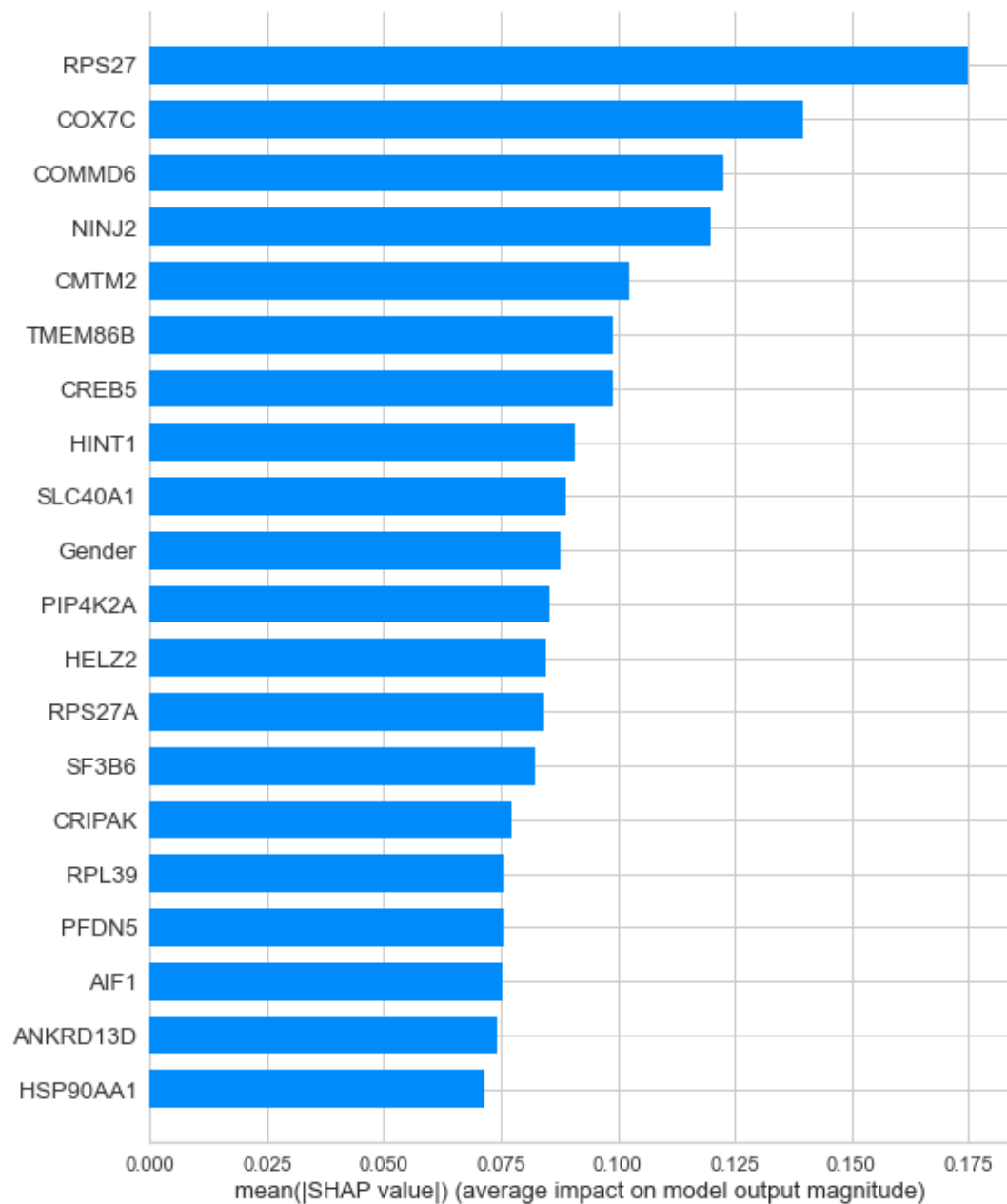


Class Prediction Error for RandomForestClassifier

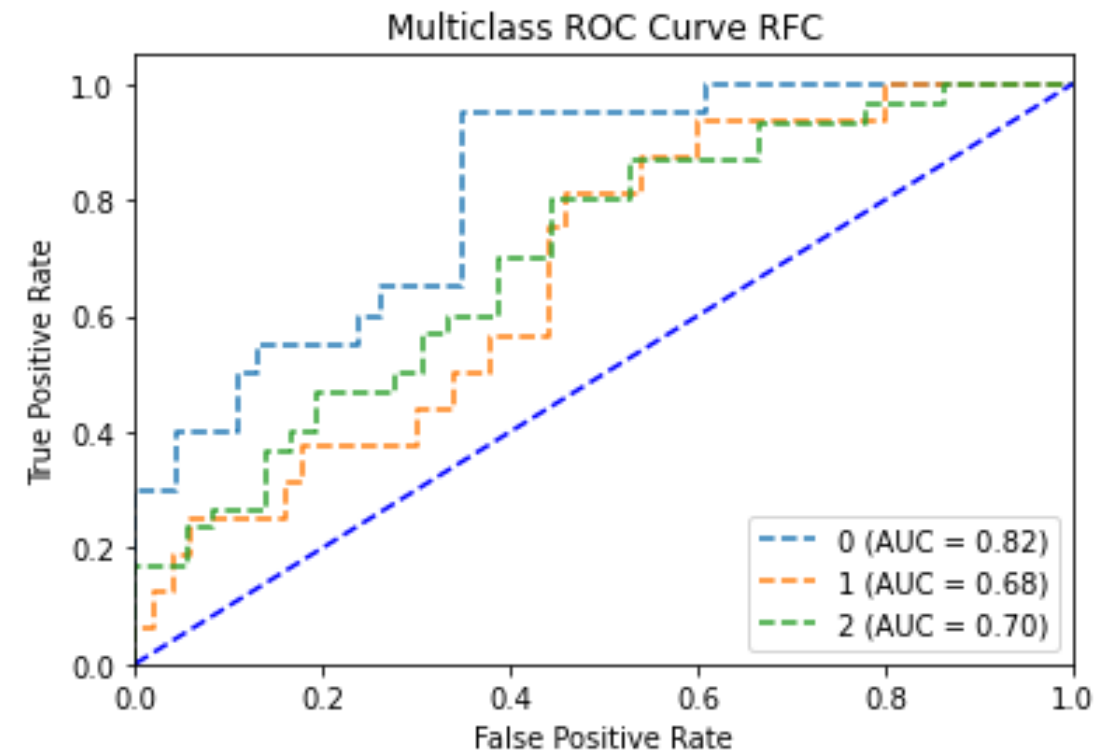
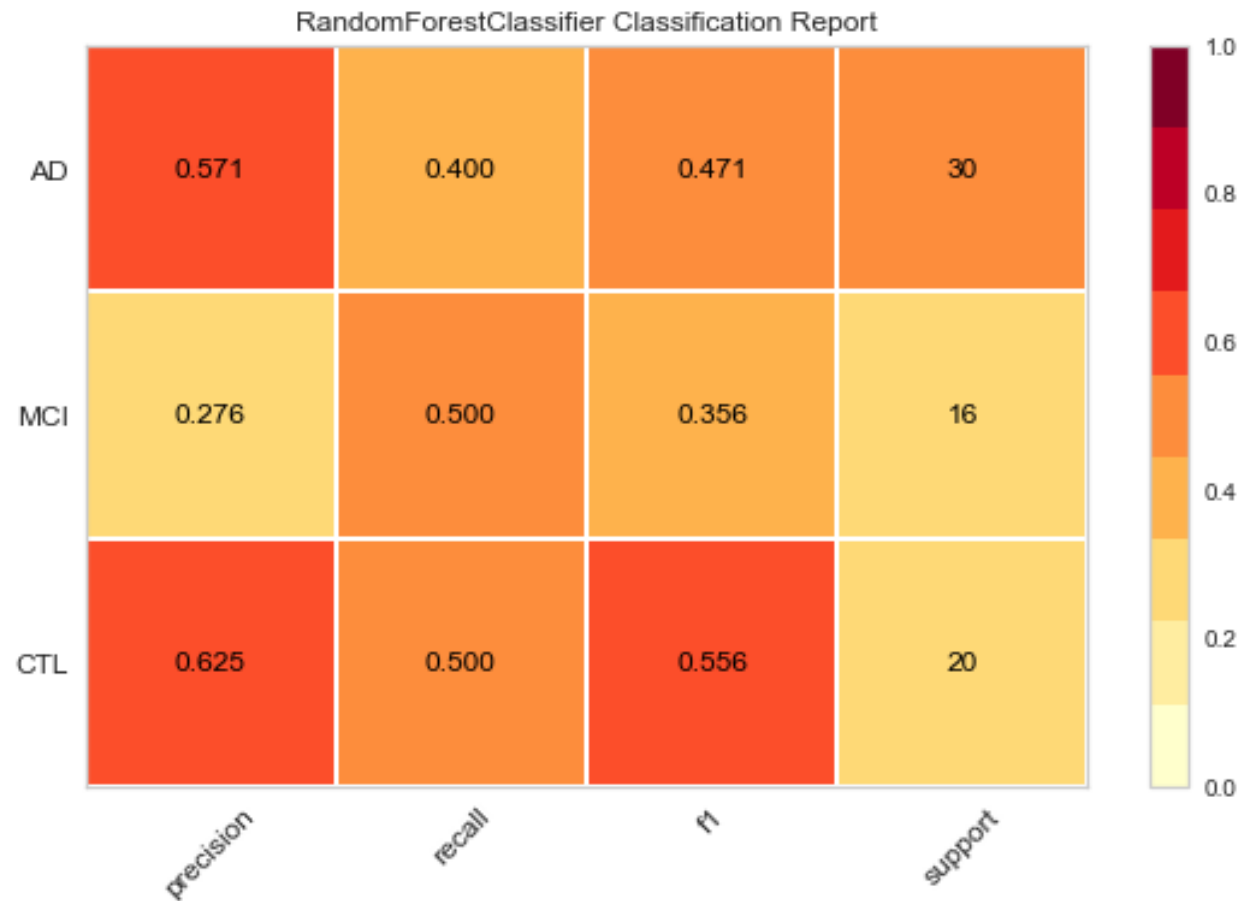


RFE_SVC selected features:-

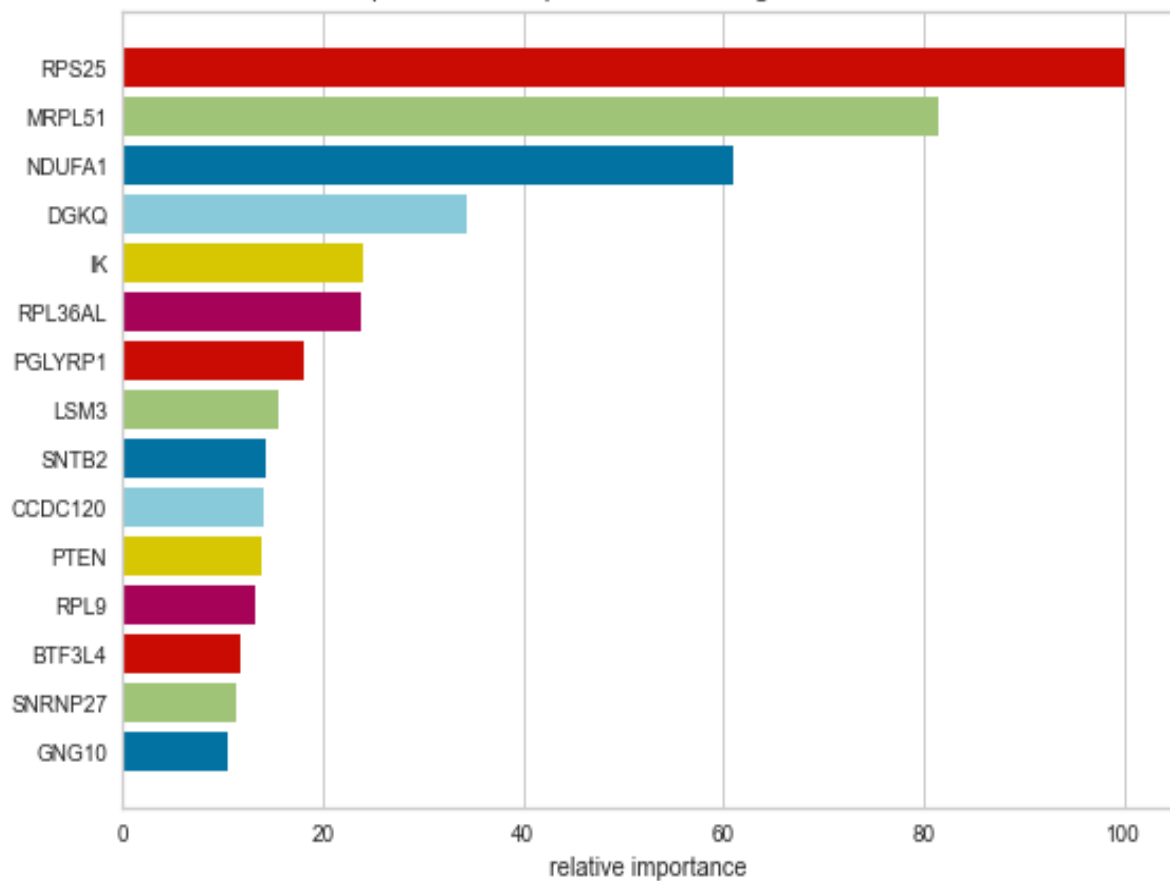




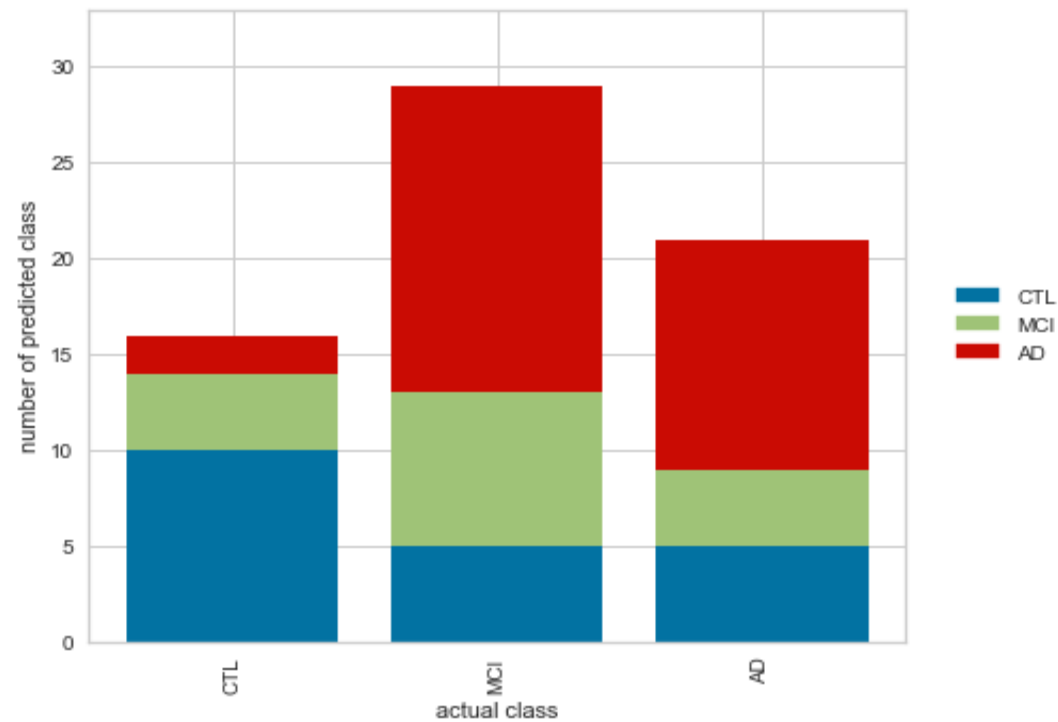
Embedded feature selection with Random Forest classifier:-



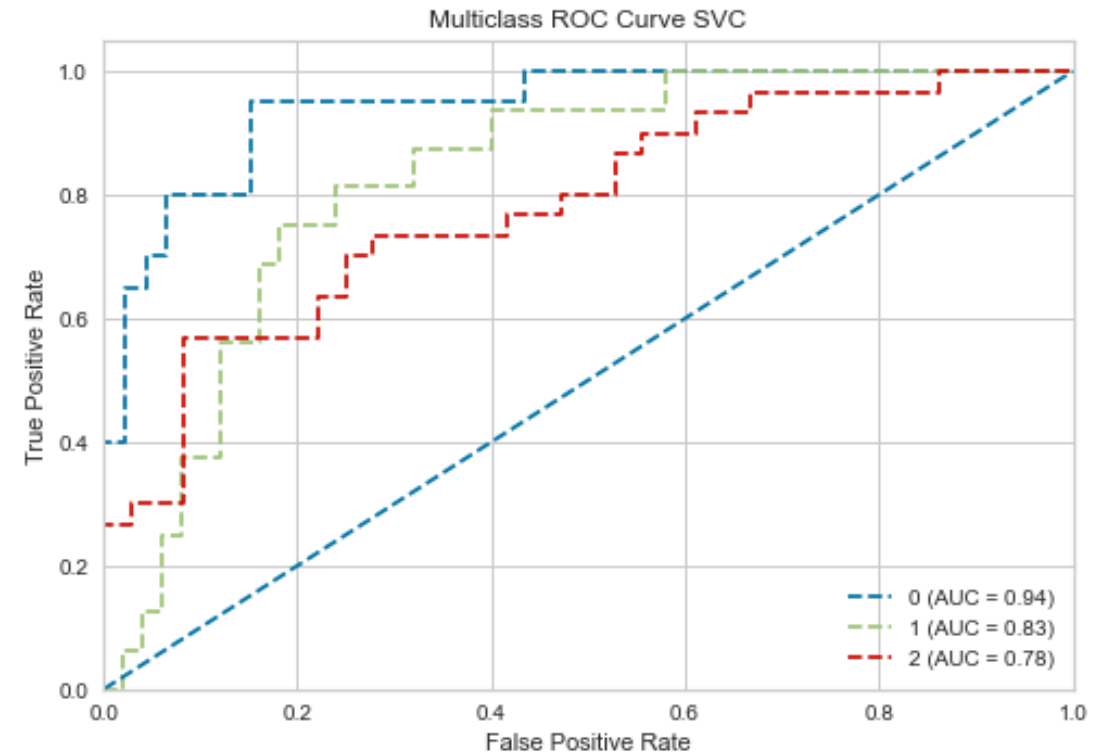
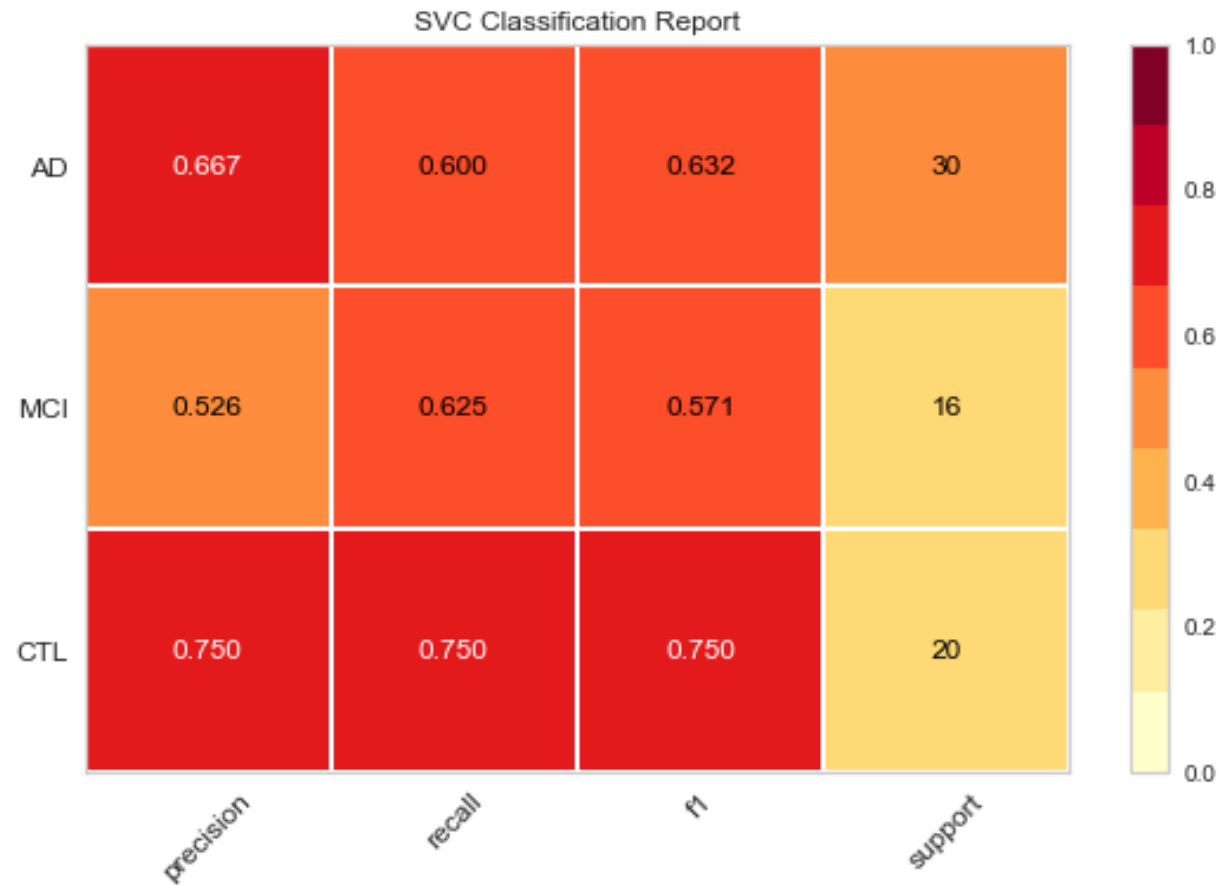
Feature Importances of Top 15 Features using RandomForestClassifier

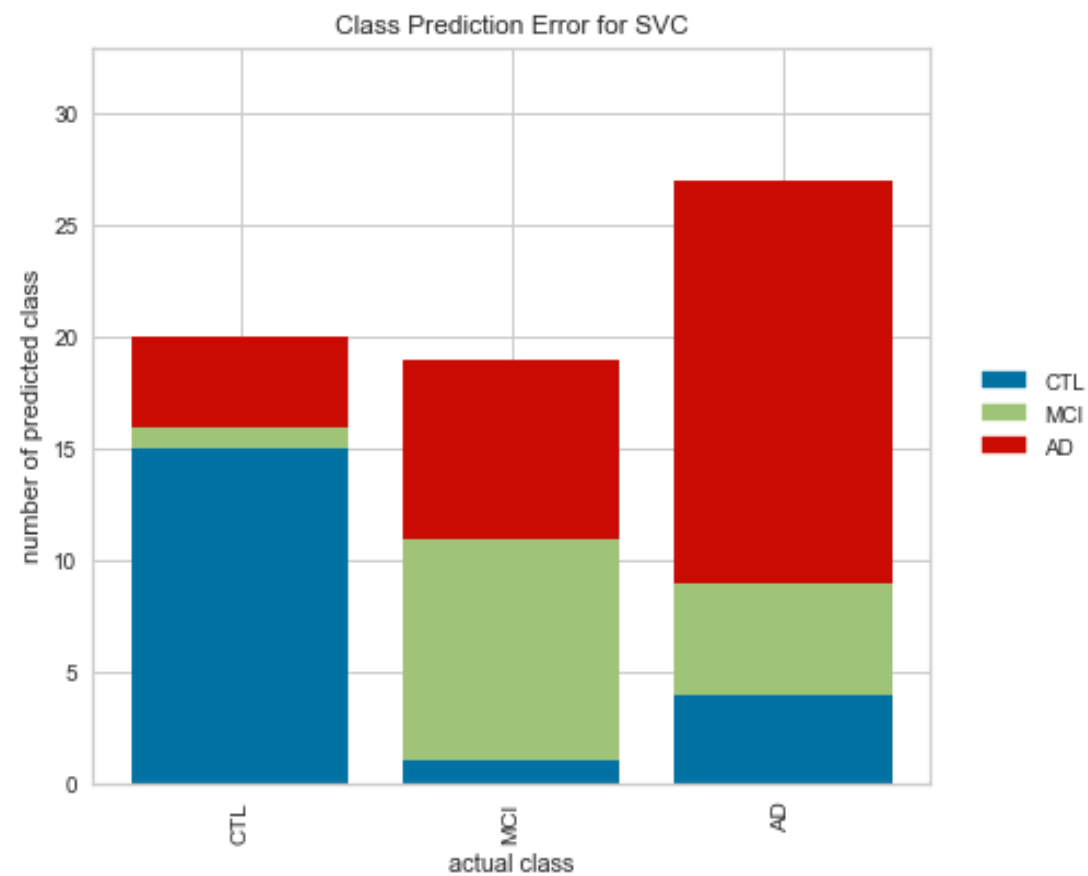
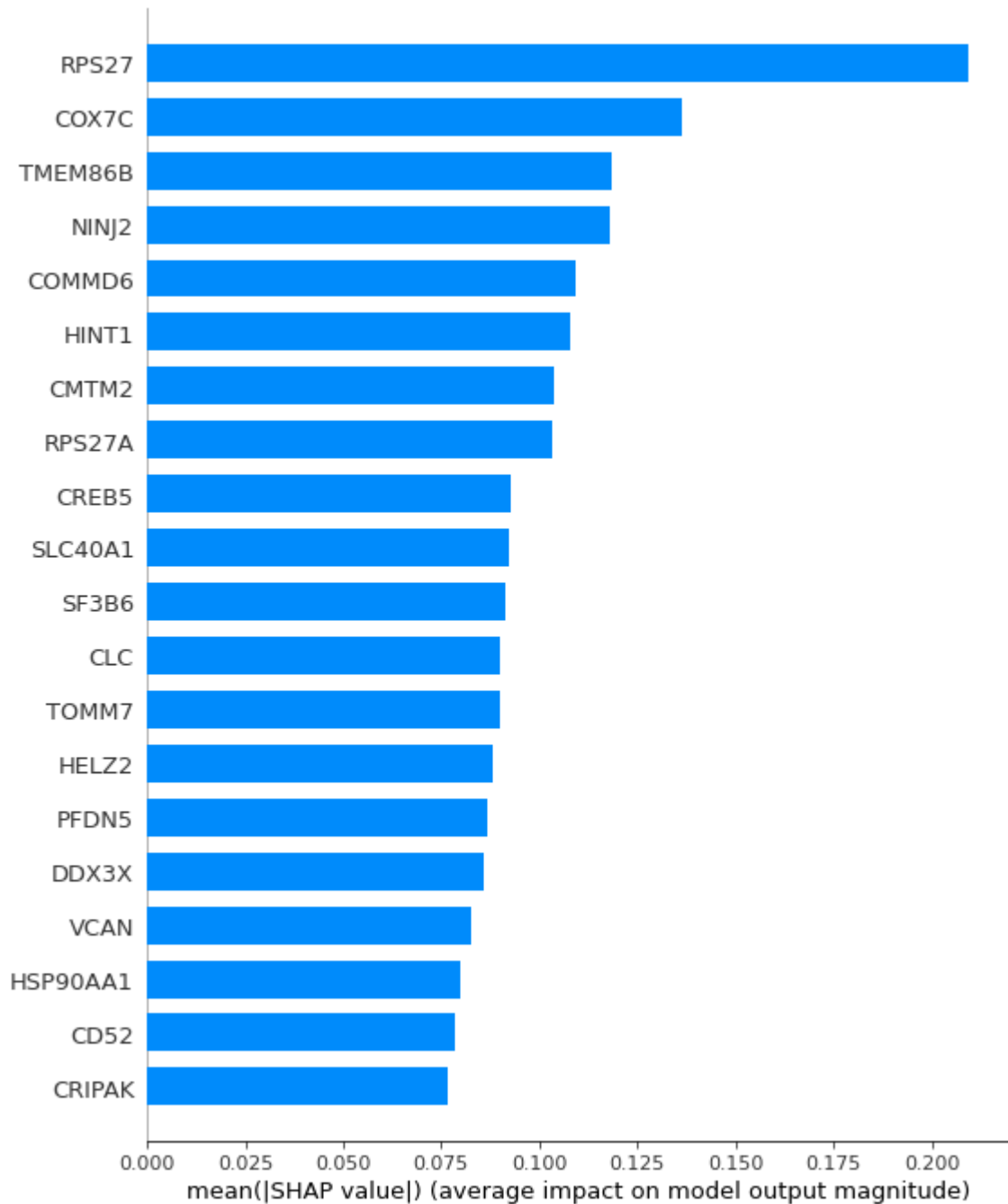


Class Prediction Error for RandomForestClassifier



Embedded feature selection with SVC:-

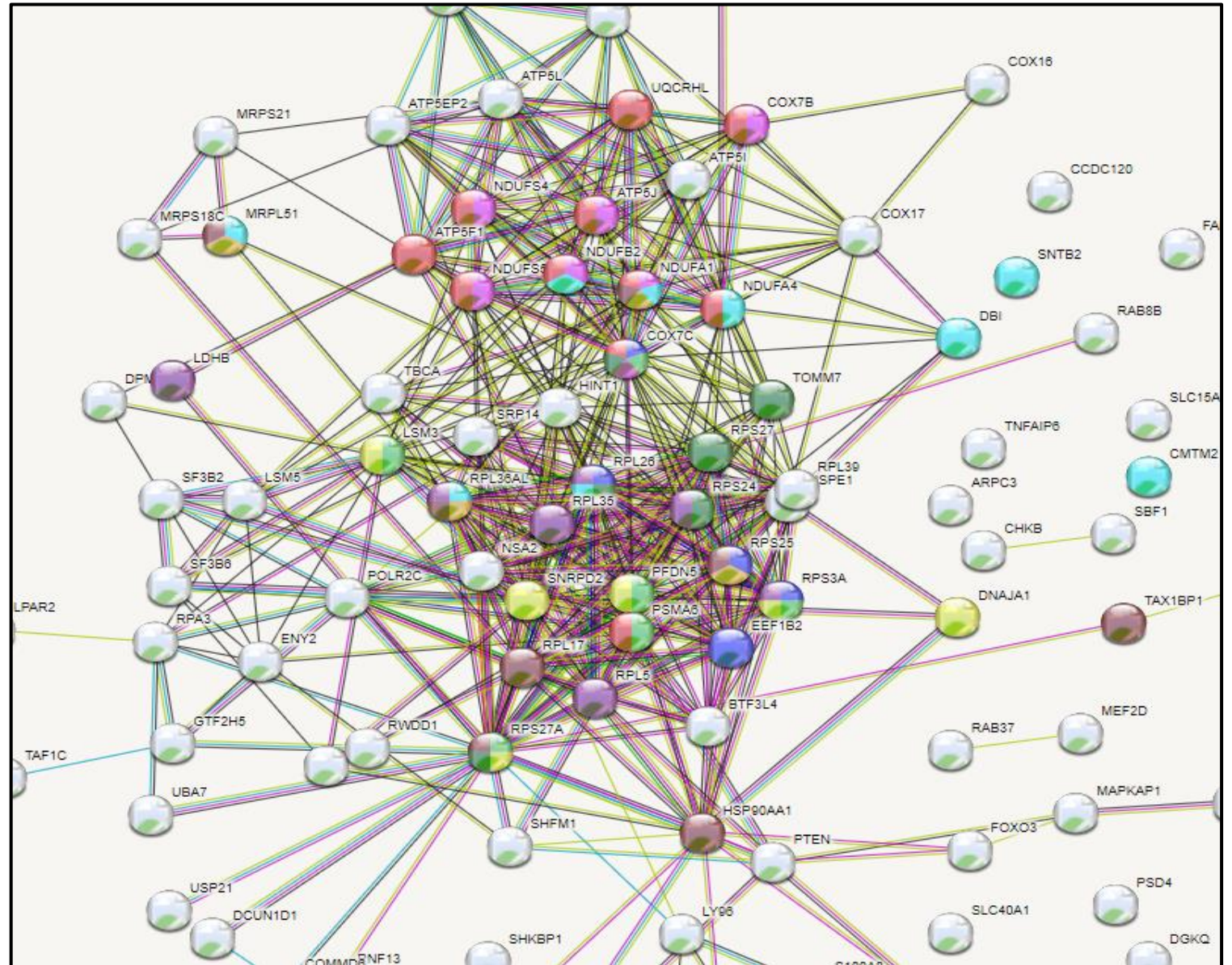




Conclusion:-

- Based on the results, genes selected by Wrapper method i.e., RFE(recursive feature elimination) with SVC(support vector classifier) are performing well in classifying three classes Healthy, MCI,AD accurately when compared with all other classifiers. As mentioned SVC classifier has accuracy of 0.75 and AUC-ROC score of 0.81 and it has comparatively less class prediction error.
- Intuition on low accuracy:-
As the number of classes increases, correctly classifying a new instance into one of them is assumed to be more challenging than making the same decision in the presence of fewer classes.

Networks obtained from
STRING of selected genes:-



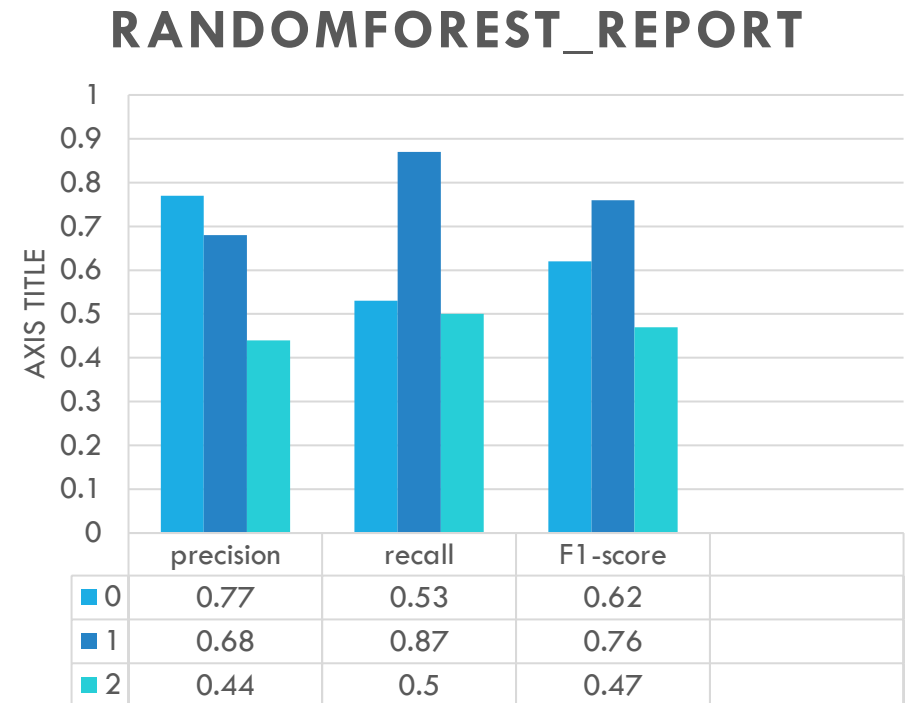
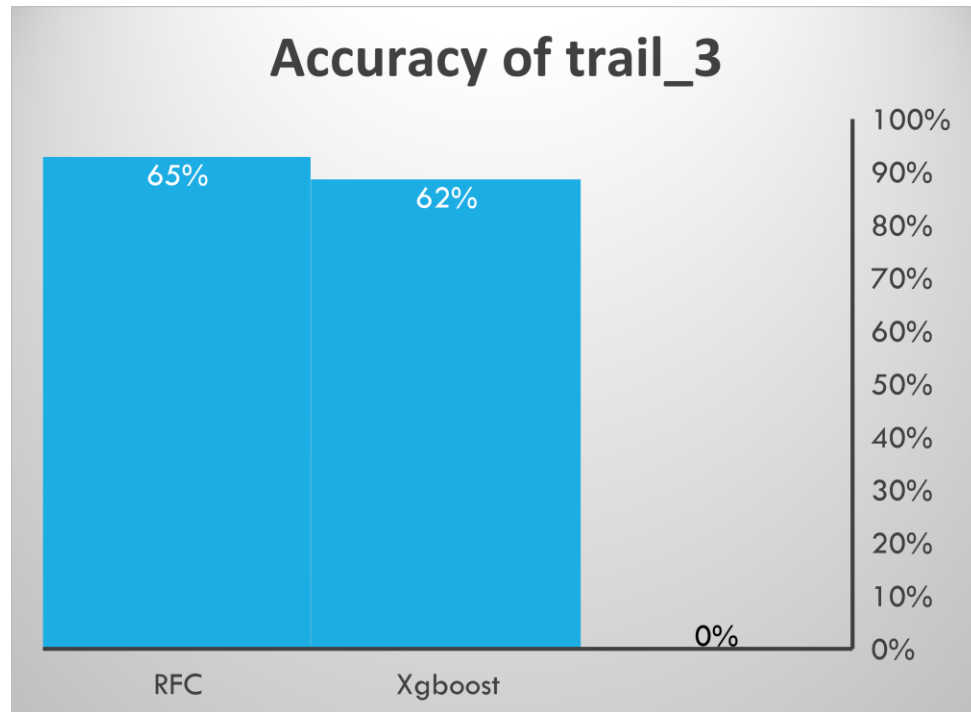
References:-

- ❖ Albert MS, DeKosky ST, Dickson D, Dubois B, Feldman HH, Fox NC, Gamst A, Holtzman DM, Jagust WJ, Petersen RC, Snyder PJ, Carrillo MC, Thies B, Phelps CH. The diagnosis of mild cognitive impairment due to Alzheimer's disease: recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimers Dement*. 2011 May;7(3):270-9. doi: 10.1016/j.jalz.2011.03.008. Epub 2011 Apr 21. PMID: 21514249; PMCID: PMC3312027.
- ❖ Shu ZY, Mao DW, Xu YY, Shao Y, Pang PP, Gong XY. Prediction of the progression from mild cognitive impairment to Alzheimer's disease using a radiomics-integrated model. *Ther Adv Neurol Disord*. 2021 Jul 15;14:17562864211029551. doi: 10.1177/17562864211029551. PMID: 34349837; PMCID: PMC8290507.
- ❖ Cui MY, Lin Y, Sheng JY, Zhang X, Cui RJ. Exercise Intervention Associated with Cognitive Improvement in Alzheimer's Disease. *Neural Plast*. 2018 Mar 11;2018:9234105. doi: 10.1155/2018/9234105. PMID: 29713339; PMCID: PMC5866875.
- ❖ Soleimani Zakeri NS, Pashazadeh S, MotieGhader H. Gene biomarker discovery at different stages of Alzheimer using gene co-expression network approach. *Sci Rep*. 2020 Jul 22;10(1):12210. doi: 10.1038/s41598-020-69249-8. PMID: 32699331; PMCID: PMC7376049.

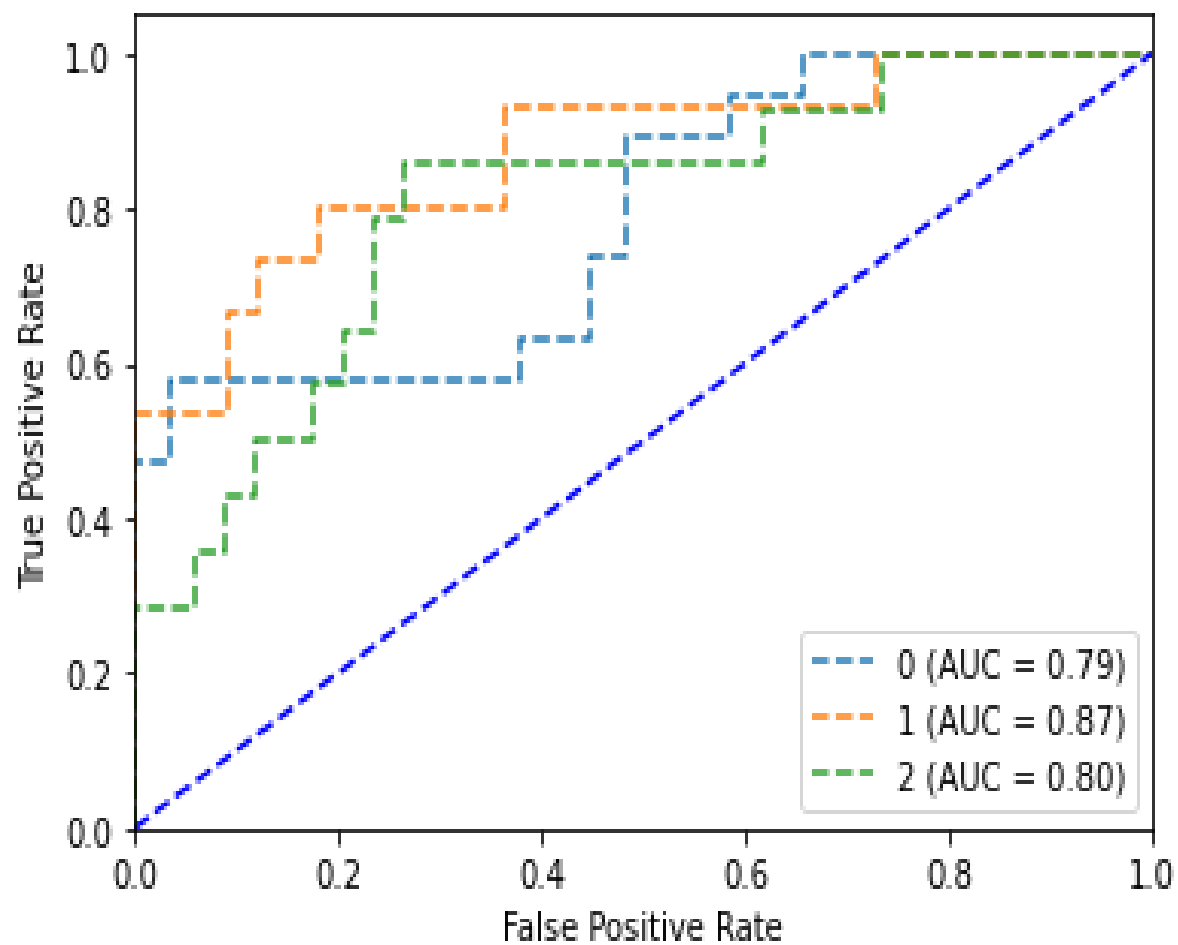
Thank you

Results of trail 3:-

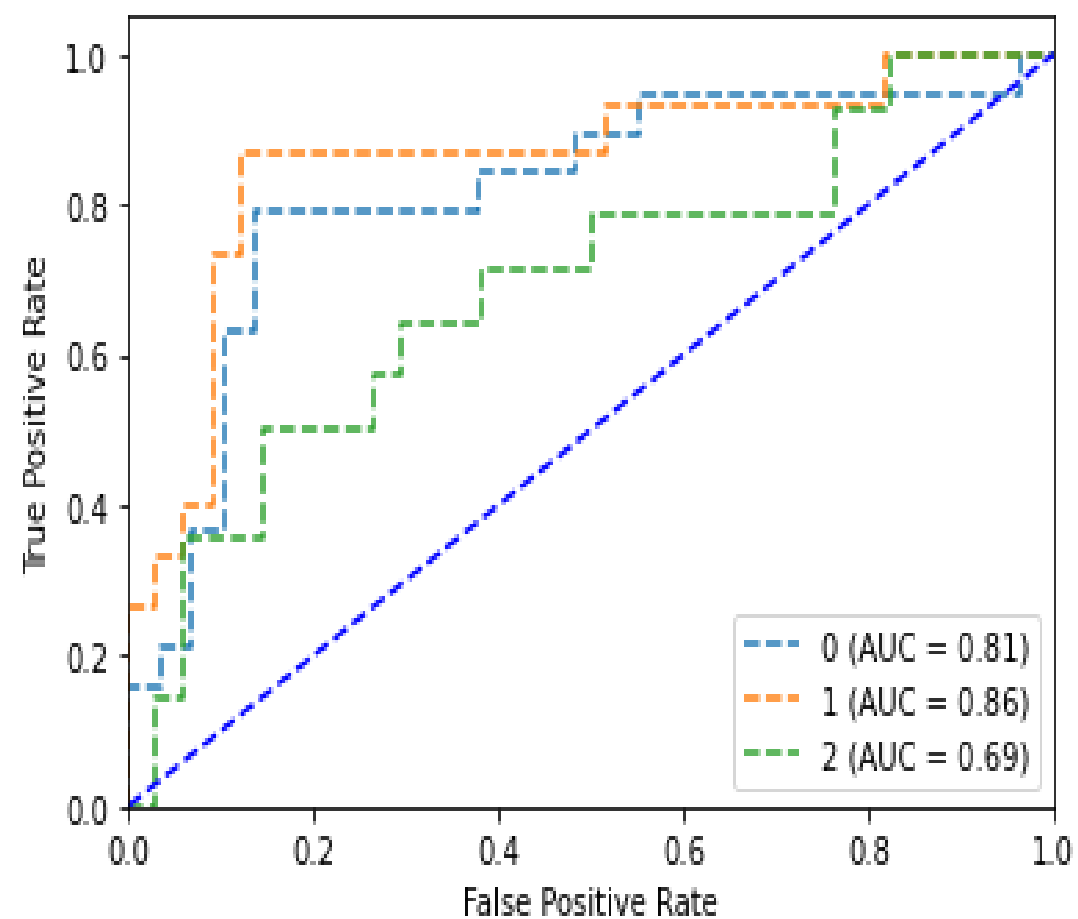
RandomForestClassifier & Xgboost classifier are performing good on the classifying target

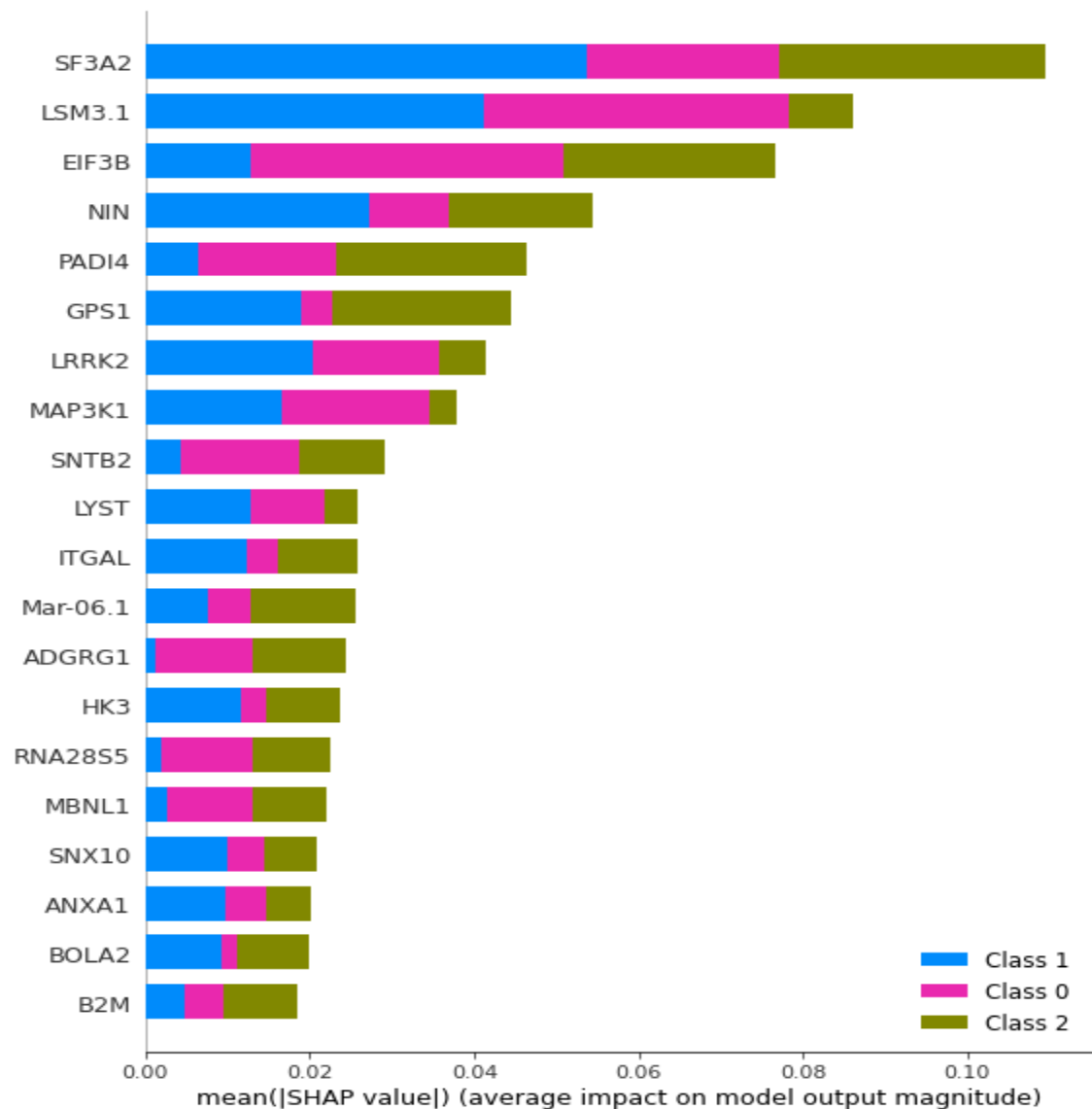
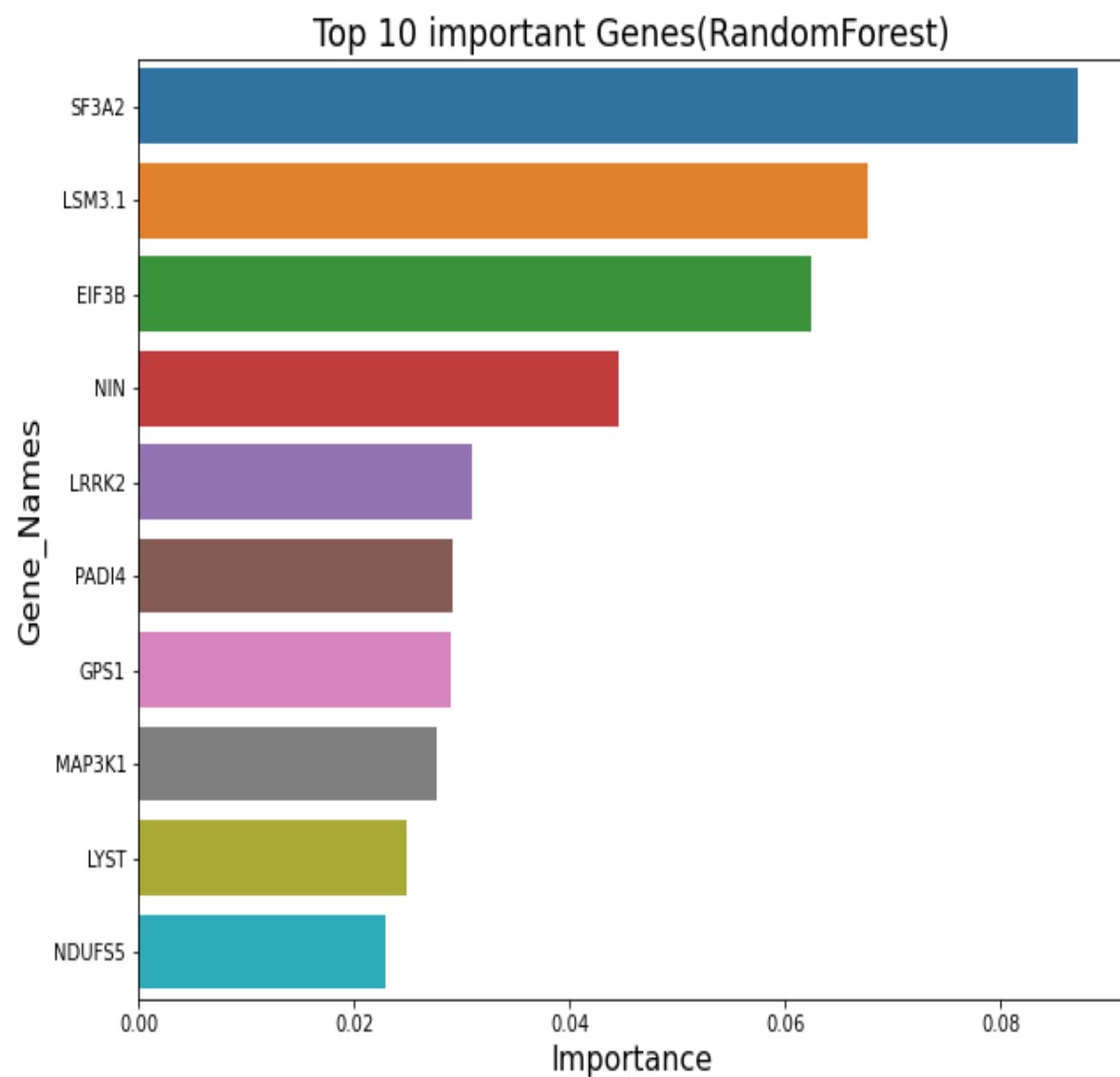


Multiclass ROC Curve RFC

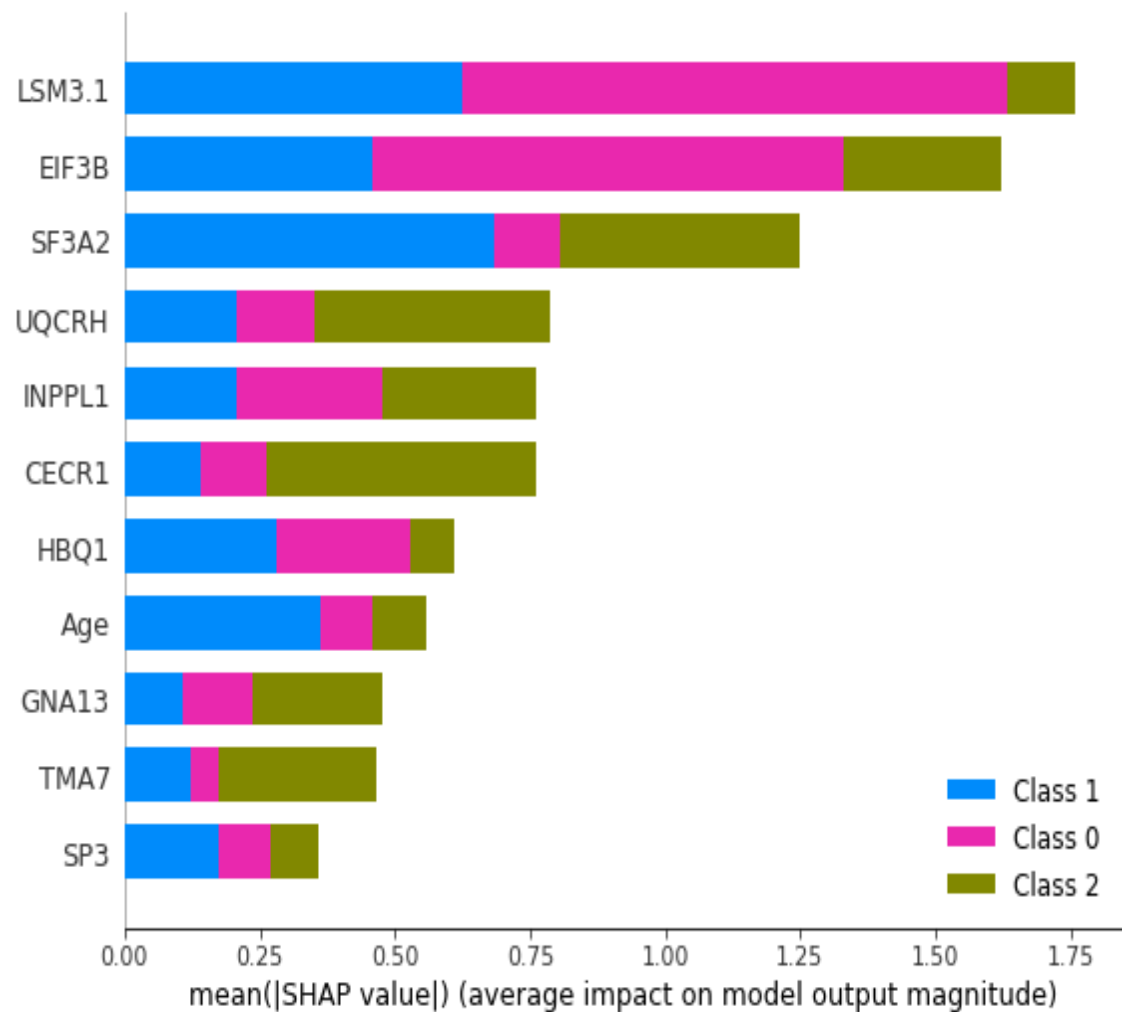
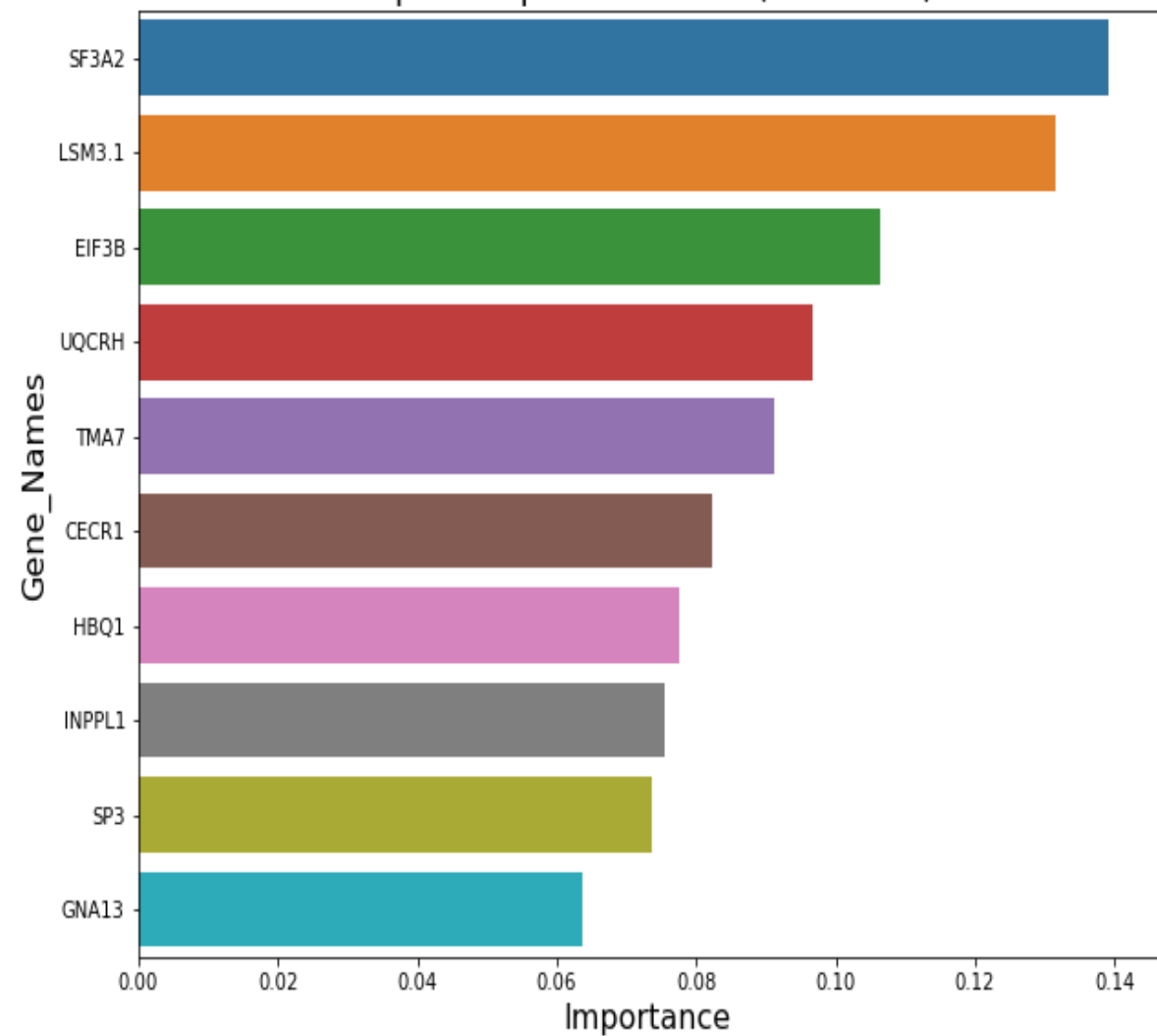


Multiclass ROC Curve XGb00st





Top 10 important Genes(XGB00ST)



Conclusion:-

Scatter plot of SF3A2 vs LSM3.1

